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Knowledge on blood donation and associated factors among residents in Ethiopia: A systematic review and Meta-analysis

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Abstract

Objective: To assess the level of knowledge regarding blood donation and associated factors in Ethiopia.

Design: systematic review and meta-analysis

Setting: This systematic review and meta-analysis was conducted in different regional states of Ethiopia in 2020.

Participants: A total of twenty articles were included in this systematic review and meta-analysis. There are 8338 study participants (4712 male & 3626 female) from different regional state of the country. Both published and unpublished cross-sectional studies were included. Articles reported other than the English language, qualitative studies, different trials, case reports, review articles, updates, news, articles not reporting the outcome of the study, and articles without full text were excluded from the analysis.

Outcome measurement: The primary outcome was level of knowledge about blood donation in Ethiopia which is measured as (adequate knowledge versus inadequate knowledge). The second outcome were factors associated with level of knowledge about blood donation in Ethiopia.

Result: The pooled prevalence of the level of knowledge about blood donation was 56.57% (95% CI: 50.30-62.84). Sex and level of education were significantly associated with the level of knowledge about blood donation. The odds of knowledge about blood donation were 3.12 times more likely among participants who had a level of education above secondary school than illiterates (AOR=3.12; 95% CI; 2.34-4.16 and 1.81 times more likely among males than females (AOR=1.81; 95% CI; 1.44-2.28).

Conclusion: The knowledge of residents about blood donation in Ethiopia was not adequate, which indicates that there is a misconception regarding blood donation in the Ethiopian population. Sex and educational status were significantly associated with the Knowledge about blood donation in Ethiopia. Therefore, there is a need for education and dissemination of information among the general population about blood donation to build adequate knowledge and maintain regular blood supply.

Keywords: *Knowledge, blood donation practice, associated factors*

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Strength and weakness of the study

- Covers wide area and accuses different articles making the review more accurate
- Subgroup and sensitivity analysis was done which overcomes heterogeneity and checking presence of influential study.
- All included studies were cross-sectional in deign which may limit to generate cause effect link.

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Introduction

Blood donation is a lifesaving practice for people who need safe blood due to different emergency conditions like road traffic accidents, surgical procedures, pregnancy and delivery complications, chemotherapy, and diseases like malaria, anemia, and intestinal parasites. Thus, the lives of these patients are depending on blood donors who can maintain an adequate supply of blood (1). Saving the lives of peoples is a noble practice and blood donation is one way to save the lives of millions. People can save up to three lives of patients by donating safe and adequate blood, which is fundamental for the survival of patients who need blood. To increase the availability of adequate and safe blood, there should be volunteer and regular blood donors to meet the increased demand for blood. Therefore, people should have adequate knowledge regarding blood donation practices to be volunteers and regular blood donors (2).

Although adequate and safe blood is in high demand, there is a serious challenge of getting blood in stocks of developing countries. Blood donation practice is highly dependent on family replacement of remunerated blood donors rather than voluntary donations of blood (3). The availability of blood for transfusion in a developing country is too limited due to the low level of knowledge, unwillingness to donate blood, and poor blood donation practice. This is a great challenge for patients who face blood loss due to different medical and surgical conditions.

Even if every blood donor is considered as a hero, the percentage of blood collected from blood donors and the average blood collection rate in developing countries including Ethiopia is extremely low as compared to the demand for blood (5). There is a great misconception regarding blood donation among residents in a different part of the world. This is due to ignorance of the population of basic knowledge regarding blood donation practices and the act of blood donation was mainly associated with replacement blood donation for those who lost a significant amount of blood during an emergency than collecting from volunteer blood donors (6).

Reports from different studies in different areas about the prevalence of knowledge regarding blood donation were low. These includes 35 % (7), 40.45 (4), 32.4% (8). Different scholars suggest that education regarding blood donation and sensitization should begin at secondary school by blood donor's organization, availing of public awareness, Compiegne through internet and media, increasing awareness and attitude, and motivational practice (9-11).

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There are different factors that hiders the knowledge of blood donation practice. These are fear of different health risks after donating blood, and lack of information on where, when, and how to donate blood (12). Lack of education and communication access regarding the demand for blood (7). There is a need to be spread awareness among the general population about blood donation to maintain adequate and safe blood supply. Although the need for blood supply is too high in Ethiopia, there is a serious shortage of blood. Thus, identifying different factors that hinder the knowledge of blood donation practice is necessary. Therefore, this systematic review and meta-analysis aim to review different studies done on knowledge of residents about blood donation in Ethiopia. Hence, the purpose of this review is to assess the knowledge of residents about blood donation and associated factors in Ethiopia.

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Method and materials

The study protocol

In this meta-analysis, Preferred Reporting Items for Systematic Review and Meta-analysis (PRISMA) guideline for reporting of the findings was used (13) (Table S1).

Databases and search strategy

The searched databases included in this review were PubMed/MEDLINE, HINARI, EMBASE, Google Scholar, and African Journal Online (AJOL). Both Published and unpublished articles from the repository of Ethiopian universities were searched. The studies were searched from January 1st, 2000 up to August 26, 2020. Articles reporting on knowledge of blood donation and associated factors among residents in Ethiopia were included in the final analysis. The search items were; "knowledge", "awareness ", "blood donation", "volunteer blood donation ", "factors", "associated factors", "determinant factors", "university students", "residents", "health care workers ", "health care providers ", "health professionals", "collage of health science students", "Ethiopia". These search strings were developed using "AND" and "OR" Boolean operators **(**Table 1**).**

Table 1: Searches from different databases about the level of knowledge and associated factors regarding blood donation among residents in Ethiopia.

Databases	Searching terms	Number of studies
MEDLINE/ PubMed	"knowledge" OR "awareness "AND "blood donation"OR "volunteer blood donation "AND "factors"OR "associated factors"OR "determinant factors"AND "university students"OR "residents"OR "health care workers "OR "health care providers "OR "health professionals"OR "collage of health science students"AND "Ethiopia".	1025
Google Scholar	"knowledge" OR "awareness "AND "blood donation"OR "volunteer blood donation "AND "associated factors"OR "determinant factors"AND "university students"OR "residents"OR "health care workers "OR "health care providers "OR "health professionals"OR "collage of health science students"AND "Ethiopia".	254
Other databases		7
Total retrieved articles		1286
Included studies		20

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Searching and eligibility of studies

All the retrieved studies were exported in to "EndNote reference software version 8 (Thomson, Stamford, CT, USA) citation manager" to sort and avoid duplication. Two investigators (AG and AD) independently evaluated each article by title and abstract using inclusion criteria, assessed the eligibility of the articles, and reviewed the selected articles critically. Extraction was done by the first name of authors, publication year, region, study setting, study subjects, sampling method, method of survey, study period, sample size, study design, level of education, professional category, level of knowledge regarding blood donation, and factors associated with the level of knowledge about blood donation.

Inclusion and exclusion criteria

Both published and unpublished cross-sectional studies conducted on knowledge of blood donation among residents in the community, health care workers and university students in Ethiopia were included. Articles reported other than the English language, qualitative studies, different trials, case reports, review articles, updates, news, articles not reporting the outcome of the study, and articles without full text were excluded from the analysis.

Outcome measurement of the study

There are two main outcomes. These are the level of knowledge on blood donation which is measured by the mean score of knowledge questions (good knowledge/poor knowledge) and factors associated with the level of knowledge about blood donation.

Quality Assessment

Two authors (AG and AD) independently assessed the quality of the studies using the Newcastle Ottawa Scale (NOS) for cross-sectional studies (14). The methodological quality of the study, comparability of the study, and the outcome and statistical analysis of the study were the three assessment tools that used to declare the quality of the study. At last, studies scored a scale of ≥ 7 out of 10 was considered as achieving high quality. During quality appraisal of the articles, any discrepancies between the two authors were resolved. All the studies were included based on the Newcastle –Ottawa Scale quality assessment criteria. All authors independently assessed the articles for consideration and inclusion in the study.

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Data processing and analysis

Prevalence regarding the level of knowledge about blood donation and factors associated were pooled using a weighted inverse variance random-effects model at 95%Cl (15). First, the data were extracted and cleaned using Microsoft Excel spreadsheets. Then the data were exported to STATA version 11 statistical software for analysis. The heterogeneity of the studies was assessed using the Cochrane Q-test and I2 with its corresponding p-value. Then the values of I2, 25%, 50%, and 75% represent low, moderate, and high heterogeneity, respectively (16). The source of heterogeneity was examined through subgroup analysis based on (region, study subjects, study setting, and sample size). Sensitivity analysis was also performed to confirm the presence or absence of influential studies. The presence of publication bias was evaluated by using Egger's test and presented with funnel plots (17). For associated factors, a log odds ratio was used to decide the association between associated factors and the level of knowledge about blood donation. A statistical test with a P value of less than 0.05 was considered statistically significant.

Patient and public involvement

It was not appropriate or possible to involve patients or the public in the design, or conduct, or reporting, or dissemination plans of our research

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Result

All the retrieved articles were 1286. Of these articles, 571 articles were excluded due to duplication. Six hundred eighty-three articles were further excluded after reviewing titles and abstracts. Of the remaining 52 articles, 32 were excluded that did not fulfill the inclusion criteria. Then, 20 articles were used for the final analysis (Figure 1).

Characteristics of the studies and study participants

A total of twenty articles were included in this systematic review and meta-analysis. These are; eight from Amhara region (1, 12, 18-23), five from Oromia region (4, 24-27), two from Addis Ababa city administration (5, 28), two from Southern Nation Nationalities and Peoples Region (SNNPR) (29, 30), two from Tigri region (31, 32), and one from Afar region (33). A total of 8338 study participants (4712 male & 3626 female) were included. All studies were cross-sectional in design and the sample size of included studies was ranged from 218 - 845 (Table 2).

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Table 2: Study characteristics included in the review and meta-analysis on knowledge about blood donation and associated factors in Ethiopia.

Author	Publication year	Region	Study participant	Method of survey	Sample size	Prevalence
Misganaw C et al.	2014	Addis Ababa	University student	Self -administered	384	83.59
Teklu S et al.	2015	Addis Ababa	Health care worker	Self –administered	295	54.24
Nigatu A & Demissie D	2014	Oromia	University student	Self -administered	399	40.35
Mulatu K et al	2017	SNNP	Community	Face to face interview	250	76.00
Gebresilase H et al.	2017	Oromia	University student	Self -administered	360	46.67
Abera B et al.	2017	Amhara	Health care worker	Self -administered	276	75.36
Mekonnen F & Melese S	2016	Amhara	Community	Face to face interview	387	56.07
Addisu AG et al.	2017	Amhara	Community	Face to face interview	376	38.30
Malako D et al.	2019	SNNP	Health care worker	Self -administered	218	82.57
Beyene GA et al.	2020	Oromia	Community	Face to face interview	410	47.07
Mijena GF et al.	2019	Oromia	Community	Self -administered	383	57.18
Jemberu YK et al.	2016	Amhara	Community	Face to face interview	772	56.48
Enawagaw B et al.	2019	Amhara	Community	Face to face interview	401	35.41
Melaku M et al.	2016	Amhara	Community	Face to face interview	768	56.77
Melaku M et al.	2018	Amhara	University student	Self -administered	255	48.24
Arage G et al.	2017	Amhara	Health care worker	Self -administered	427	51.99
Urgesa K et al.	2017	Oromia	Community	Face to face interview	845	43.55
Tadesse T et al.	2018	Tigri	Health care worker	Self -administered	556	57.19
Tadesse W et al.	2018	Afar	University student	Self -administered	339	53.98
Seid T et al.	2017	Tigri	Health care worker	Self -administered	237	70.46

SNNP, South Nation Nationalities and Peoples

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Knowledge about blood donation

The overall pooled prevalence regarding knowledge about blood donation in Ethiopia was 56.57% (95% CI: 50.30-62.84) (Figure 2).

Heterogeneity and publication bias

In this systematic review and meta-analysis, heterogeneity was detected within the studies (I2 = 97.3%, p<0.001). The funnel plot showed that there is an asymmetrical distribution of studies included in the review and Egger's test was statistically significant (p=0.02), suggesting the presence of publication bias (Figure 3).

Subgroup analysis

In this systematic review and meta-analysis, subgroup analysis was done by region, study subjects, study setting, and sample size. The highest pooled prevalence of knowledge about blood donation was reported on health care workers, 65.28% (95% CI: 55.16, 75.40). Similarly, the level of knowledge about blood donation was higher among studies done in institutions (institution-based) 60.45% (95% CI: 51.05, 69.83) than studies done in the community. The prevalence of knowledge was reported among those studies with a sample size of <400, 60.25% (95% CI: 51.20, 6931) (Table 3).

Table 3: Subgroup analysis on level of knowledge about blood donation and associated factors in Ethiopia (n=20)

Variables	Subgroup	Studies (n)	Population	Prevalence (95%CI)	I^2 (%)	P-value
Region	Northern Ethiopia	10	4455	54.81 (47.65, 61.57)	95.7	<0.001
	Southern Ethiopia	5	1637	58.52 (41.66,76.38)	98.2	<0.001
	Other	5	2246	58.53(42.67,74.38)	98.4	<0.001
Study participants	Health care workers	6	2009	65.28(55.16, 75.40)	95.9	<0.001
	Community	9	4592	51.84 (44.60,59.08)	96.1	<0.001
	University students	5	1737	54.62 (36.86,72.42)	95.8	<0.001
Study setting	Institution based	11	3746	60.45 (51.05, 69.83)	97.5	<0.001
	Community based	9	4592	51.84 (44.80,59.08)	96.1	<0.001
Sample size	<400	13	4159	60.25 (51.20, 6931)	97.6	<0.001
	≥ 400	7	4179	49.84 (43.90, 55.79)	93.5	<0.001

Other: Addis Ababa, Eastern Ethiopia

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Sensitivity analysis

As reviled in table 4, none of the point's estimates outside of the overall 95% confidence interval, which confirms that there is no influential study (Table 4).

Table 4: Sensitivity analysis of knowledge about blood donation and associated factors in Ethiopia

Study omitted	Estimate prevalence	(95%CI)
Misganaw C et al.(2014)	55.12	(49.57,60.67)
Teklu S et al.(2015)	56.69	(50.15,63.24)
Nigatu A and Demissie DB (2014)	57.42	(51.04, 63.80)
Mulatu K et al(2017)	55.55	(49.24, 6186)
Gebresilase HW et al.(2017)	57.09	(50.58, 63.59)
Abera B et al.(2017	55.58	(49.27, 61.90)
Mekonnen FH and Melese ST(2016)	56.60	(50.01,63.18)
Addisu AG et al.(2017)	57.53	(51.20, 63.86)
Malako D et al.(2019)	55.21	(49.14, 61.27)
Beyene GA et al.(2020)	57.07	(50.55, 63.59)
Mijena GF et al.(2019)	56.54	(49.95, 63.12)
Jemberu YK et al.(2016)	56.57	(49.84, 63.31)
Enawagaw B et al. (2019)	57.69	(51.48, 63.90)
Melaku M et al.(2016)	56.56	(49.83, 63.29)
Melaku M et al.(2018)	57.00	(50.50, 63.50)
Arage G et al.(2017)	56.81	(50.23, 63.39)
Urgesa K et al.(2017)	57.27	(50.82, 63.72)
Tadesse T et al.(2018)	56.54	(49.89, 63.19)
Tadesse W et al.(2018)	56.71	(50.14, 62.28)
Seid T et al.(2017)	55.85	(49.41, 62.28)
Overall	56.57	(50.30, 62.84)

CI: Confidence Interval

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Factors associated with knowledge on blood donation

This study showed that there is a significant association between sex and knowledge about blood donation. The odds of knowledge were 1.81times more likely among males than females (AOR=1.81; 95% CI; 1.44-2.28) (Figure 4). This review also showed that there is an association between level of education and level of knowledge about blood donation. The odds of knowledge were 3.12 times more likely among participants whose educational status was above secondary school than those who did not attend any formal education (AOR=3.12; 95% CI; 2.34-4.16) (Figure 5).

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Discussion

In this systematic review and meta-analysis, the pooled prevalence of knowledge about blood donation among residents in Ethiopia was 56.57% (95% CI: 50.30-62.84). This indicates that there is an awareness problem with blood donation in the country. This might be due to different reasons like unable to incorporate blood donation programs in formal education, less media coverage regarding information about blood donation, limited Compiegne regarding blood donation, and limited educational access in the country.

The result of this study is lower than studies conducted in Malaysia (98%) (2), Barabanki, India (90%) (34), and Benin City, Nigeria (92.65) (35). This variation might be due to the difference of the study subjects in which the population of interest in the previous studies was; nursing students in Malaysia, health care workers in Benin City, Nigeria, undergraduate medical students in Barabanki India, but in the current study, the population included residents in the community, health care workers, and university students. It might be also due to the variation of media coverage and educational status of the people's sine Ethiopia is one of the low-income countries having inadequacies regarding education and media accessibility. There is almost a similar finding regarding knowledge of blood donation in a study done among students of a tertiary institution in Nigeria (9). However, the prevalence of knowledge about blood donation in this study is higher than studies conducted in Kerala, India (7), and Nepal (8) in which the prevalence of knowledge was 35% and 32% respectively. It may be due to the time variation in which the previous studies were conducted before seven and five years, respectively.

In subgroup analysis, there is no significant variation regarding the prevalence of knowledge about blood donation within a region where the studies are done. However, there is a variation of knowledge prevalence within study participants, study setting, and sample size category. Accordingly, the highest prevalence of knowledge was reported among health care workers. This is because health care workers had formal education regarding blood donation and know more about human physiology than other parts of population. Studies conducted in institutions (institution-based) had a higher prevalence of knowledge than studies conducted in the community (community-based). This is due to the reason that populations found in the institutions had more information access (7).

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In this systematic review and meta-analysis, the odds of knowledge about blood donation were 1.81 times more likely among males than females. This might be due to the reason that females are more prone to different physiological conditions like menstruation, bleeding during pregnancy, and delivery which deters them to donate blood. As a result, they will have poor blood donation practices when compared with males. This poor practice may indirectly affect a female's knowledge about blood donation (3). Educational status had also a significant association with the knowledge of blood donation. The odds of knowledge about blood donation were 3.12 times more likely among participants who had a level of education above secondary school than illiterates (11). This is because formal education is a cornerstone for acquiring knowledge. Therefore, participants having an educational status of above secondary level have information accesses than participants who did not have any formal education.

Conclusion

The knowledge of residents about blood donation in Ethiopia was not adequate, which indicates that there is a misconception regarding blood donation in the Ethiopian population. Sex and educational status were significantly associated with the Knowledge about blood donation in Ethiopia. Therefore, there is a need for education and dissemination of information among the general population about blood donation to build adequate knowledge and maintain regular blood supply.

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Abbreviations

AJO: African Journals Online, AOR: Adjusted Odds Ratio, CI: Confidence Interval NOS, Newcastle Ottawa Scale, PRISMA: Preferred Reporting Items for Systematic Review and Meta-analysis, and SNNP: South Nation Nationalities and Peoples.

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Declarations

Availability of data and materials

All related data have been presented within the manuscript. The dataset supporting the conclusions of this article is available from the authors on request.

Author's Contributions

AG and MB designed the study, designed and run the literature search. All authors (AG, AW, MB, GG and AD) acquired data, screened records, extracted data, and assessed risk of bias. AG and AD did statistical analyses and wrote the report. All authors provided critical conceptual input, analyzed and interpreted data, and critically revised the report. All authors read and approved the final manuscript.

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Ethics approval and consent to participate

Not applicable

Consent for publication

Not applicable.

Competing interest

The authors declared that they have no competing interests.

Acknowledgment

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Figure legends

1. Figure.1: Flow chart of selection for systematic review and meta-analysis on the level of knowledge about blood donation and associated factors in Ethiopia.
2. Figure 2: Forest plot of the pooled prevalence of knowledge about blood donation and associated factors in Ethiopia.
3. Figure 3: Funnel plot with 95% confidence limits of the pooled prevalence of knowledge about blood donation and associated factors in Ethiopia.
4. Figure 4: The overall pooled odds ratio of the association between sex and level of knowledge about blood donation in Ethiopia.
5. Figure 5**:** The overall pooled odds ratio of the association between educational status and level of knowledge about blood donation in Ethiopia

Additional file: Table S1: Preferred Reporting Items for Systematic Review and Meta-analysis guideline for reporting of the findings

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List of figures

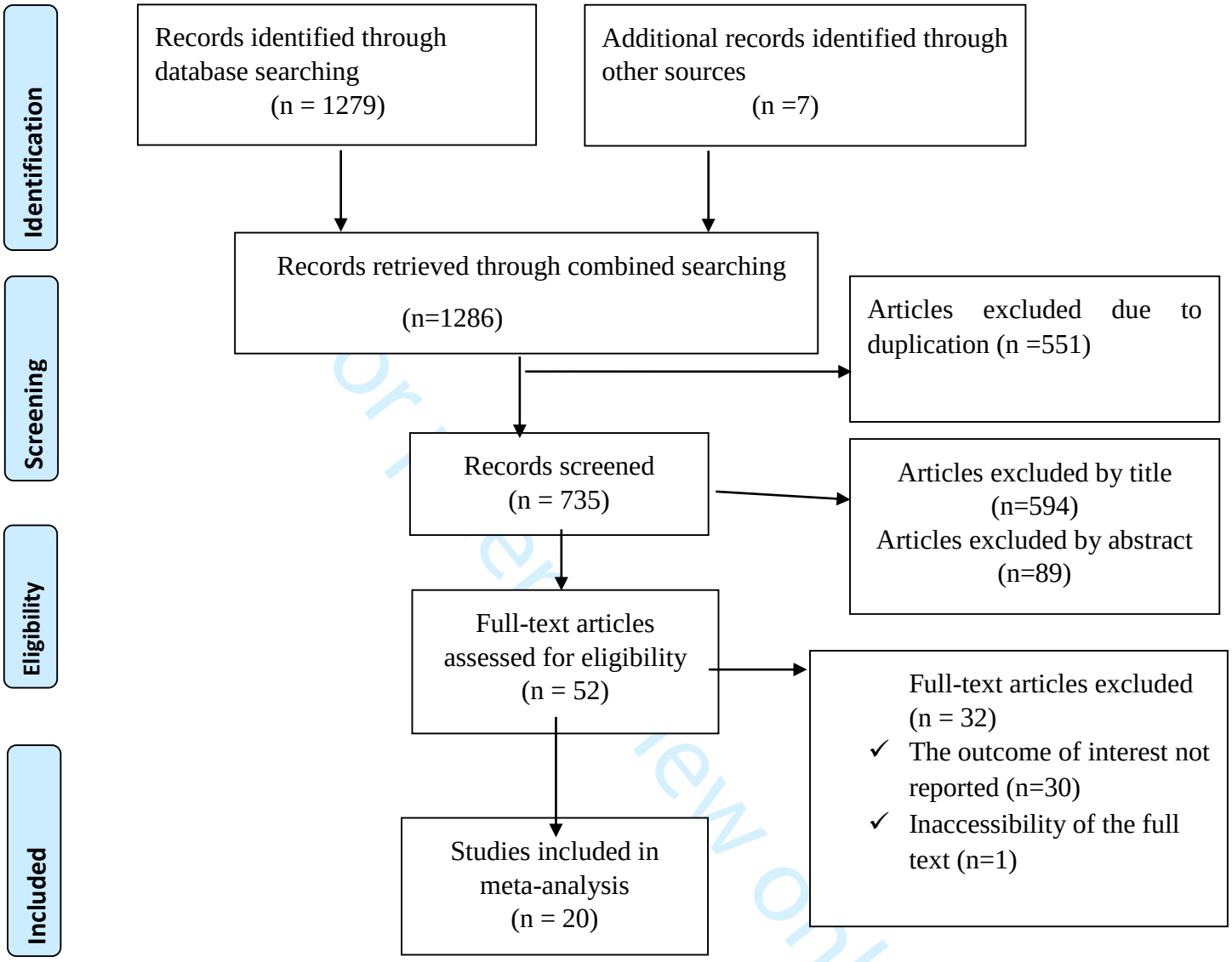


Figure.1: Flow chart of selection for systematic review and meta-analysis on the level of knowledge about blood donation and associated factors in Ethiopia.

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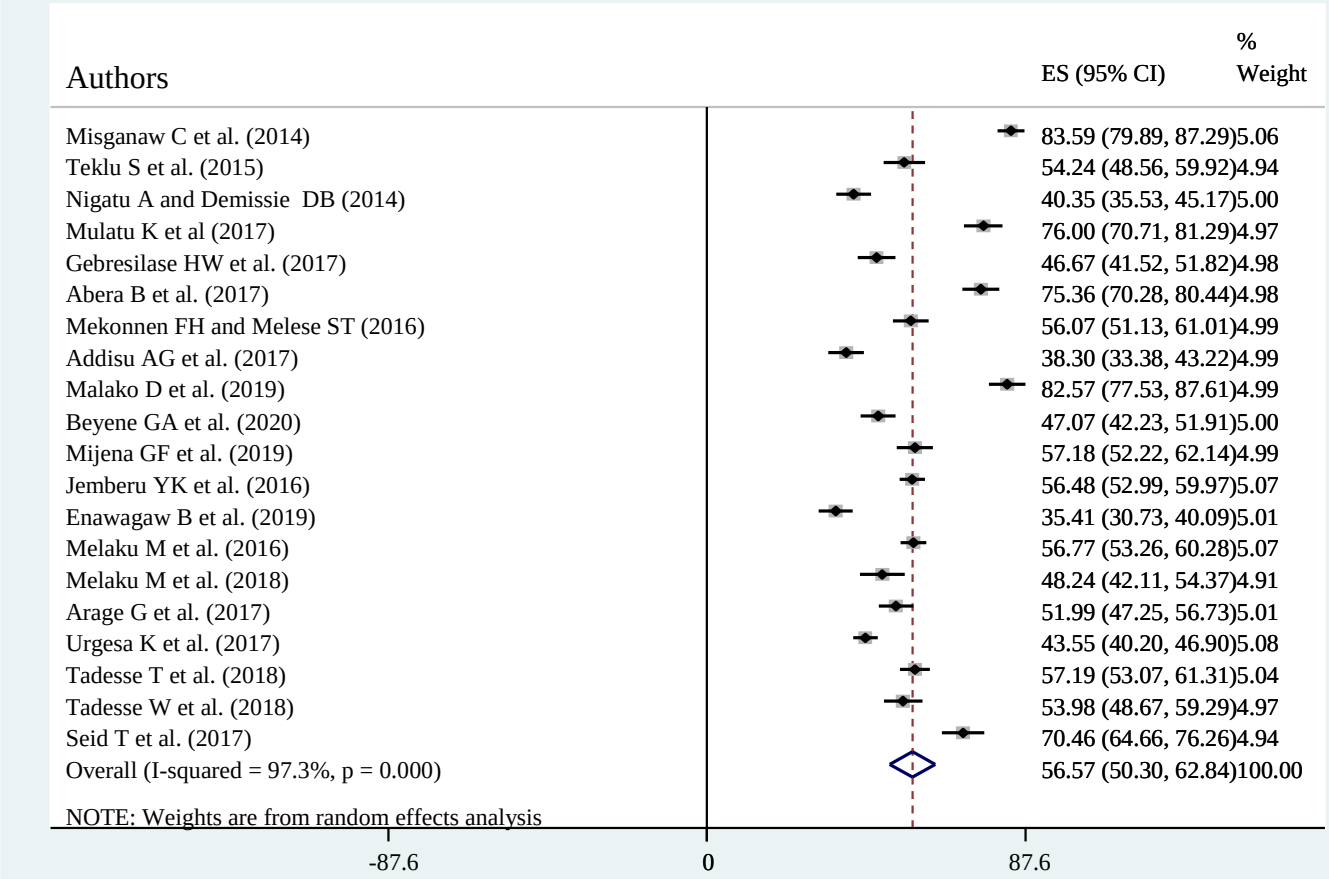


Figure 2: Forest plot of the pooled prevalence of knowledge about blood donation and associated factors in Ethiopia.

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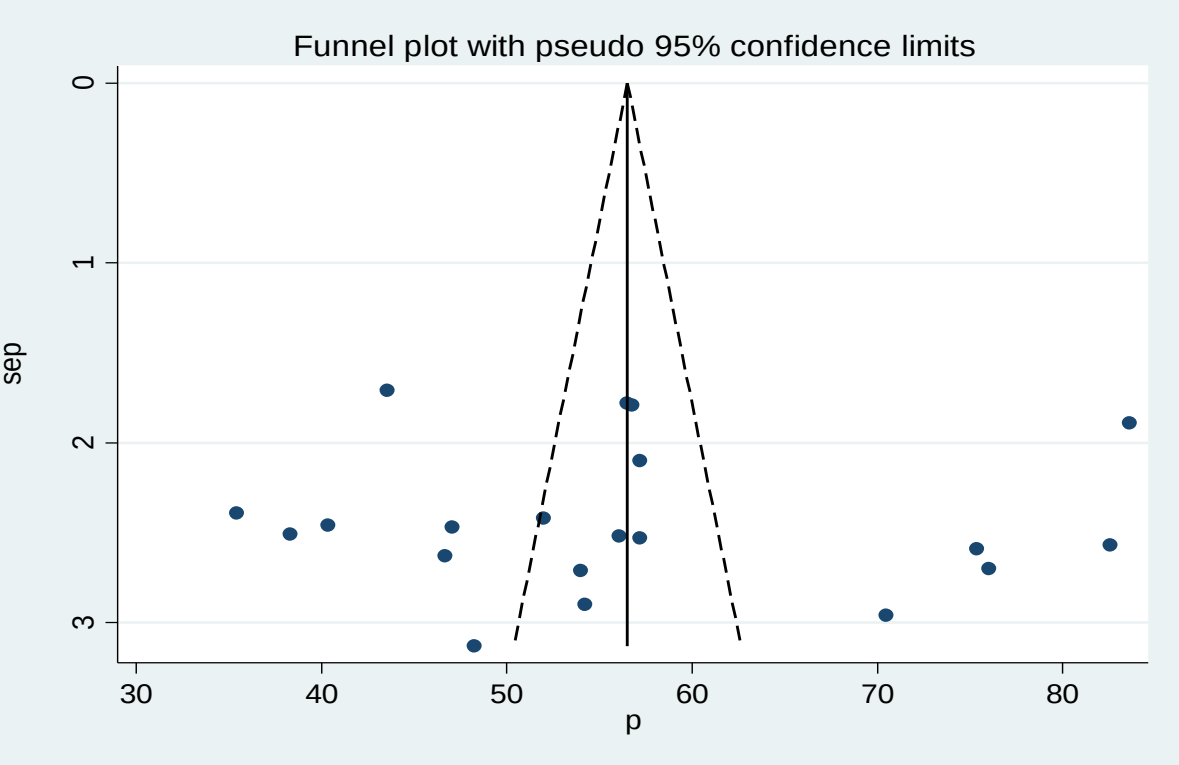


Figure 3: Funnel plot with 95% confidence limits of the pooled prevalence of knowledge about blood donation and associated factors in Ethiopia.

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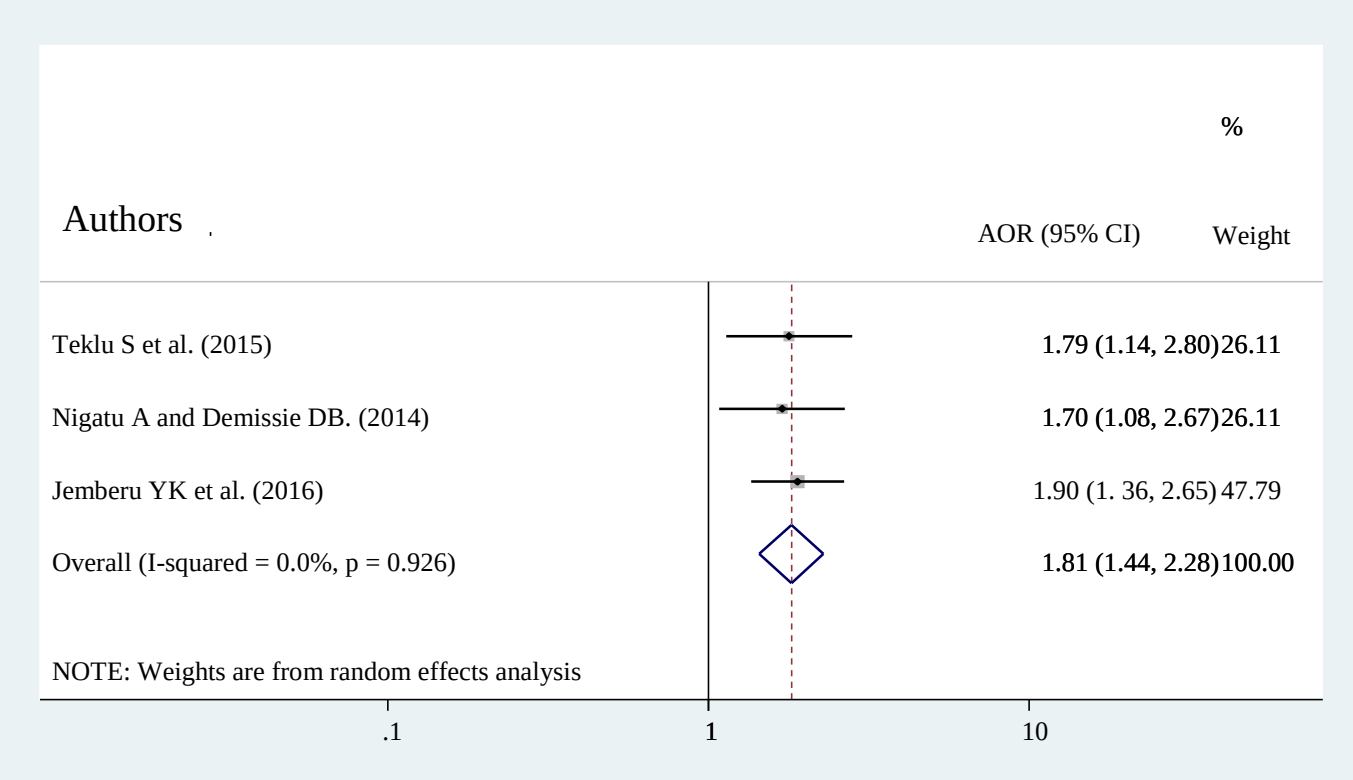


Figure 4: The overall pooled odds ratio of the association between sex and level of knowledge about blood donation in Ethiopia.

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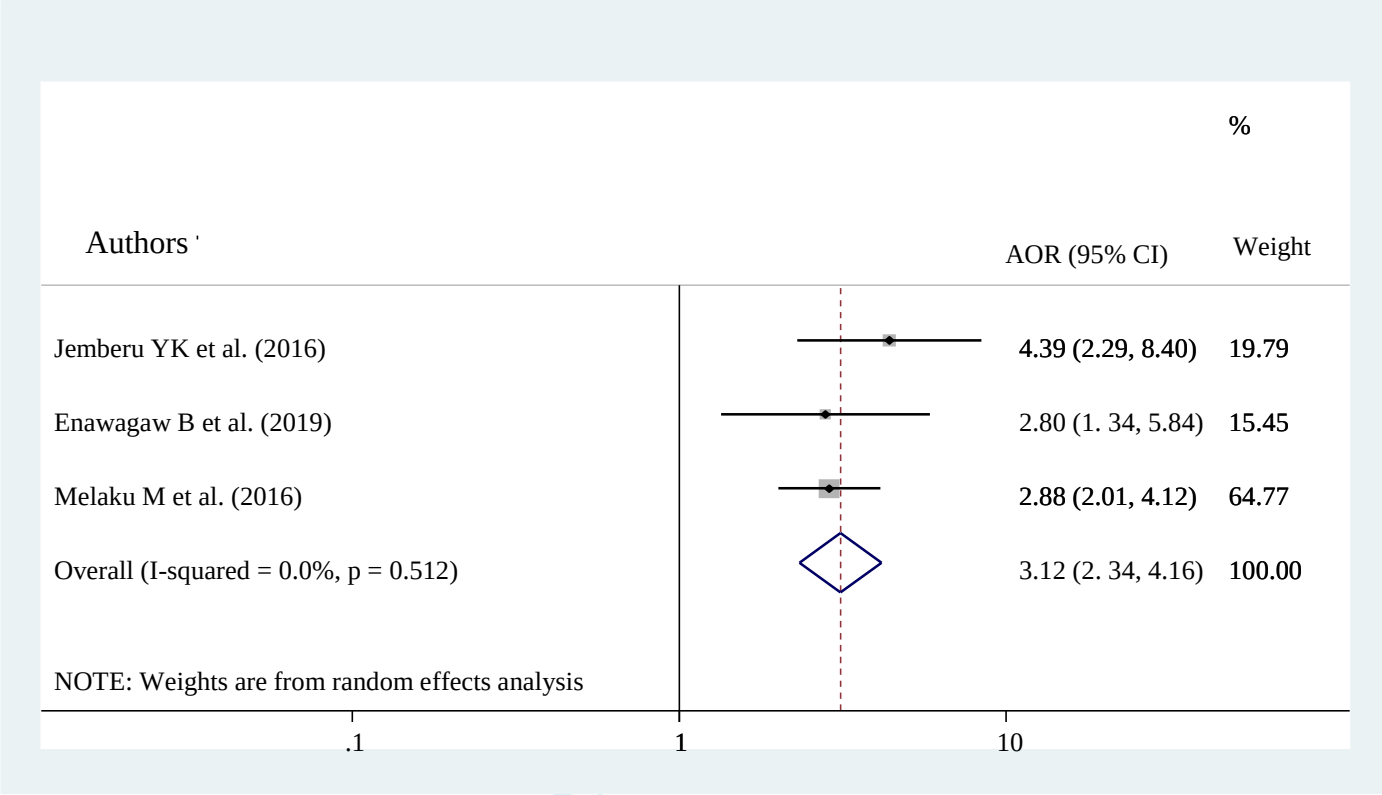


Figure 5**:** The overall pooled odds ratio of the association between educational status and level of knowledge about blood donation in Ethiopia.

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PRISMA 2009 Checklist

Table 1: Preferred Reporting Items for Systematic Review and Meta-analysis guideline for reporting of the findings

Section/topic	#	Checklist item	Reported on page #
TITLE			1
Title	1	Identify the report as a systematic review, meta-analysis, or both.	1
ABSTRACT			2
Structured summary	2	Provide a structured summary including, as applicable: background; objectives; data sources; study eligibility criteria, participants, and interventions; study appraisal and synthesis methods; results; limitations; conclusions and implications of key findings; systematic review registration number.	2
INTRODUCTION			3
Rationale	3	Describe the rationale for the review in the context of what is already known.	3
Objectives	4	Provide an explicit statement of questions being addressed with reference to participants, interventions, comparisons, outcomes, and study design (PICOS).	3
METHODS			4
Protocol and registration	5	Indicate if a review protocol exists, if and where it can be accessed (e.g., Web address), and, if available, provide registration information including registration number.	NA
Eligibility criteria	6	Specify study characteristics (e.g., PICOS, length of follow-up) and report characteristics (e.g., years considered, language, publication status) used as criteria for eligibility, giving rationale.	5
Information sources	7	Describe all information sources (e.g., databases with dates of coverage, contact with study authors to identify additional studies) in the search and date last searched.	4
Search	8	Present full electronic search strategy for at least one database, including any limits used, such that it could be repeated.	4
Study selection	9	State the process for selecting studies (i.e., screening, eligibility, included in systematic review, and, if applicable, included in the meta-analysis).	5
Data collection process	10	Describe method of data extraction from reports (e.g., piloted forms, independently, in duplicate) and any processes for obtaining and confirming data from investigators.	5
Data items	11	List and define all variables for which data were sought (e.g., PICOS, funding sources) and any assumptions and simplifications made.	5
Risk of bias in individual studies	12	Describe methods used for assessing risk of bias of individual studies (including specification of whether this was done at the study or outcome level), and how this information is to be used in any data synthesis.	6
Summary measures	13	State the principal summary measures (e.g., risk ratio, difference in means).	6
Synthesis of results	14	Describe the methods of handling data and combining results of studies, if done, including measures of consistency (e.g., I^2) for each meta-analysis.	6

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PRISMA 2009 Checklist

Section/topic	#	Checklist item	Reported on page #
Risk of bias across studies	15	Specify any assessment of risk of bias that may affect the cumulative evidence (e.g., publication bias, selective reporting within studies).	6
Additional analyses	16	Describe methods of additional analyses (e.g., sensitivity or subgroup analyses, meta-regression), if done, indicating which were pre-specified.	6
RESULTS			7
Study selection	17	Give numbers of studies screened, assessed for eligibility, and included in the review, with reasons for exclusions at each stage, ideally with a flow diagram.	7
Study characteristics	18	For each study, present characteristics for which data were extracted (e.g., study size, PICOS, follow-up period) and provide the citations.	8
Risk of bias within studies	19	Present data on risk of bias of each study and, if available, any outcome level assessment (see item 12).	10
Results of individual studies	20	For all outcomes considered (benefits or harms), present, for each study: (a) simple summary data for each intervention group (b) effect estimates and confidence intervals, ideally with a forest plot.	9
Synthesis of results	21	Present results of each meta-analysis done, including confidence intervals and measures of consistency.	10
Risk of bias across studies	22	Present results of any assessment of risk of bias across studies (see Item 15).	11
Additional analysis	23	Give results of additional analyses, if done (e.g., sensitivity or subgroup analyses, meta-regression [see Item 16]).	12
DISCUSSION			14
Summary of evidence	24	Summarize the main findings including the strength of evidence for each main outcome; consider their relevance to key groups (e.g., healthcare providers, users, and policy makers).	14
Limitations	25	Discuss limitations at study and outcome level (e.g., risk of bias), and at review-level (e.g., incomplete retrieval of identified research, reporting bias).	16
Conclusions	26	Provide a general interpretation of the results in the context of other evidence, and implications for future research.	16
FUNDING			NA
Funding	27	Describe sources of funding for the systematic review and other support (e.g., supply of data); role of funders for the systematic review.	NA

From: Moher D, Liberati A, Tetzlaff J, Altman DG, The PRISMA Group (2009). Preferred Reporting Items for Systematic Reviews and Meta-Analyses: The PRISMA Statement. PLoS Med 6(7): e1000097. doi:10.1371/journal.pmed1000097

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Knowledge on blood donation and associated factors in Ethiopia: A systematic review and Meta-analysis

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Abstract

Objective: To assess the level of knowledge on blood donation and associated factors in Ethiopia.

Design: Systematic review and meta-analysis

Methods: Both published and unpublished cross-sectional studies conducted on knowledge of blood donation in Ethiopia were included. Articles reported other than the English language, qualitative studies, trials, case reports, review articles, updates, news, articles not reporting the outcome of the study, and articles without full text were excluded from the analysis. PubMed/MEDLINE, HINARI, EMBASE, Scopus, Google Scholar, and African Journals Online (AJOL) were searched for the study. Cochrane I^2 statistics were used to check the presence of heterogeneity. Subgroup analysis and sensitivity analysis were done on evidence of heterogeneity. Egger test with funnel plot was conducted to investigate publication bias.

Result: Twenty cross-sectional studies were included for the analysis of this systematic review and meta-analysis. A total of 8,338 study participants (4,712 males & 3,626 females) were involved. The overall pooled level of knowledge about blood donation was 56.57% (95% CI: 50.30-62.84). Having a secondary and above level of education (AOR=3.12; 95% CI; 2.34-4.16) and being male (AOR=1.81; 95% CI; 1.44-2.28) were factors associated with blood donation.

Conclusion: More than half of the study participants were knowledgeable about blood donation. Sex and educational status were significantly associated with knowledge about blood donation in Ethiopia. Therefore, there is a need for education and dissemination of information among the general population about blood donation to build adequate knowledge and maintain a regular blood supply.

Keywords: *Knowledge, blood donation practice, associated factors*

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Strength and weakness of the study

- Covers a wide area and accuses different articles making the review more accurate
- Subgroup analysis and sensitivity analysis were done on evidence of heterogeneity.
- All included studies were cross-sectional in design, which may limit the generation of a cause-effect link.

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Introduction

Blood donation is a lifesaving practice for people who need safe blood due to different emergency conditions like road traffic accidents, surgical procedures, pregnancy and delivery complications, chemotherapy, and diseases like malaria, anemia, and intestinal parasites. Thus, the lives of these patients are depending on blood donors who can maintain an adequate supply of blood (1). Saving the lives of peoples is a noble practice and blood donation is one way to save the lives of millions. People can save up to three lives of patients by donating safe and adequate blood, which is fundamental for the survival of patients who need blood. To increase the availability of adequate and safe blood, there should be volunteer and regular blood donors to meet the increased demand for blood. Therefore, people should have adequate knowledge regarding blood donation practices to be volunteers and regular blood donors (2).

Although adequate and safe blood is in high demand, there is a serious challenge of getting blood in stocks from developing countries. Blood donation practice is depending on the family replacement of remunerated blood donors. In most developing and transitional countries, family replacement and paid blood donors are a significant source of blood transfusion (3). The availability of blood for transfusion in a developing country is too limited due to the low level of knowledge, unwillingness to donate blood, and poor blood donation practice. This is a great challenge for patients who face blood loss due to different medical and surgical conditions (4).

Even if every blood donor is considered a hero, the amount of blood collected from blood donors and the average blood collection rate in developing countries including Ethiopia is low compared to the demand for blood due to increased major operations and infectious diseases like HIV/AIDS (5). There is a great misconception regarding blood donation in different part of the world. This is due to ignorance of the population of basic knowledge regarding blood donation practices and the act of blood donation was mainly associated with replacement blood donation for those who lost a significant amount of blood during an emergency than collected from volunteer blood donors (6).

Reports from different studies in different areas about the prevalence of knowledge regarding blood donation were low. These includes 35% (7), 40.45% (4), 32.4% (8). Different scholars suggest that education regarding blood donation and sensitization should begin at secondary school by blood donor's organization, availing of public awareness, Compiegne through internet and media, increasing awareness and attitude, and motivational practice (9-11).

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Different factors hinder the knowledge of blood donation practice. These are the fear of different health risks after donating blood, and lack of information on where, when, and how to donate blood (12). Even though beliefs, attitudes, and behaviors regarding blood donation differ, more knowledge is needed among non-donors. Fear is the commonest deterrent to blood donation, whereas altruism is the commonest motivating factor to donate blood. There is a gap between willingness to blood donation and the number of donors in most populations around the world, which affects blood donation (13). Communication strategies must be designated to overcome these issues leading to the act of blood donation. Lack of education and communication access regarding the demand for blood (7). There is a need to spread awareness among the general population about blood donation to maintain an adequate and safe blood supply. Although the need for blood supply is too high in Ethiopia, there is a serious shortage of blood. Thus, identifying different factors that hinder the knowledge of blood donation practice is necessary. Therefore, this systematic review and meta-analysis aim to review different studies done on the level of knowledge about blood donation and associated factors in Ethiopia.

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Methods and materials

Study protocol

In this meta-analysis, the Preferred Reporting Items for Systematic Review and Meta-analysis (PRISMA) guideline for reporting the findings was used (14) (Table S1).

Databases and search strategy

The searched databases included in this review were PubMed/MEDLINE, HINARI, EMBASE, Google Scholar, and African Journal Online (AJOL). Both Published and unpublished articles from the repository of Ethiopian universities were searched. The studies were searched from January 1st, 2000 up to August 26, 2020. Articles reporting on knowledge of blood donation and associated factors in Ethiopia were included in the final analysis. The search items were; "knowledge", "awareness ", "blood donation", "volunteer blood donation ", "factors", "associated factors", "determinant factors", "university students", "residents", "health care workers ", "health professionals", "collage of health science students", "Ethiopia". These search strings were developed using "AND" and "OR" Boolean operators **(**Table 1**).**

Table 1: Searches from different databases about the level of knowledge and associated factors regarding blood donation in Ethiopia.

Databases	Searching terms	Number of studies
MEDLINE/ PubMed	"knowledge" OR "awareness "AND "blood donation" OR "volunteer blood donation "AND "factors" OR "associated factors" OR "determinant factors" AND "university students" OR "residents" OR "health care workers "OR "health care providers "OR "health professionals" OR "collage of health science students" AND "Ethiopia".	1025
Google Scholar	"knowledge" OR "awareness "AND "blood donation" OR "volunteer blood donation "AND "associated factors" OR "determinant factors" AND "university students" OR "residents" OR "health care workers "OR "health care providers "OR "health professionals" OR "collage of health science students" AND "Ethiopia".	254
Other databases		7
Total retrieved articles		1286
Included studies		20

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Searching and eligibility of studies

All retrieved studies were exported into "EndNote reference software version 8 (Thomson, Stamford, CT, USA) citation manager" to sort and avoid duplication. Two investigators (AG and AD) independently evaluated each article by title and abstract using inclusion criteria, assessed the eligibility of the articles, and reviewed the selected articles critically. Extraction was done by the first name of authors, publication year, region, study setting, study subjects, sampling method, method of survey, study period, sample size, study design, level of education, professional category, level of knowledge regarding blood donation, and factors associated with the level of knowledge about blood donation.

Eligibility criteria

Both published and unpublished cross-sectional studies conducted on knowledge of blood donation among populations in the community, health care workers, and university students in Ethiopia were included, whereas qualitative studies, different trials, case reports, review articles, updates, and news were excluded from the analysis. Furthermore, articles not reporting the outcome of the study and those without full text were excluded.

Outcome measurement of the study

There are two main outcomes. These are the level of knowledge on blood donation and factors associated with the level of knowledge about blood donation. The level of knowledge was measured by using the mean score of knowledge questions. Thus, those study participants who scored mean and above on the knowledge assessment items were considered as knowledgeable, whereas those who scored below the mean score were considered as not knowledgeable about blood donation.

Quality Assessment

Two authors (AG and AD) independently assessed the quality of the studies using the Newcastle Ottawa Scale (NOS) for cross-sectional studies (15). The methodological quality of the study, comparability of the study, and the outcome and statistical analysis of the study were the three assessment tools that were used to declare the quality of the study. At last, studies that scored a scale of ≥ 7 out of 10 were considered as achieving high quality. During quality appraisal of the articles, any discrepancies between the two authors were resolved. All studies were included based on the Newcastle –Ottawa Scale quality assessment criteria. All authors independently assessed the articles for consideration and inclusion in the study.

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Data processing and analysis

Prevalence regarding the level of knowledge about blood donation and factors associated were pooled using a weighted inverse variance random-effects model at 95%Cl (16). First, the data were extracted and cleaned using Microsoft Excel spreadsheets. Then the data were exported to STATA version 11 statistical software for analysis. The heterogeneity of the studies was assessed using the Cochrane Q-test and I2 with its corresponding p-value. Then the values of I^2: 25%, 50%, and 75% represent low, moderate, and high heterogeneity, respectively (17). The source of heterogeneity was examined through subgroup analysis based on (region, study subjects, study setting, and sample size). Sensitivity analysis was also performed to confirm the presence or absence of influential studies. The presence of publication bias was evaluated by using Egger's test and presented with funnel plots (18). For associated factors, a log odds ratio was used to decide the association between associated factors and the level of knowledge about blood donation. A statistical test with a P value of less than 0.05 was considered statistically significant.

Patient and public involvement

It was not appropriate or possible to involve patients or the public in the design, conduct, reporting, or dissemination plans of our research

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Result

The retrieved articles were 1286. Of these articles, 551 articles were excluded due to duplication. Six hundred eighty-three articles were further excluded after reviewing titles and abstracts. Of the remaining 52 articles, 32 were excluded which did not fulfill the inclusion criteria. Then, 20 articles were used for the final analysis (Figure 1).

Characteristics of the studies and study participants

A total of twenty articles were included in this systematic review and meta-analysis. These are; eight from the Amhara region (1, 12, 19-24), five from the Oromia region (4, 25-28), two from Addis Ababa city administration (5, 29), two from Southern Nation Nationalities and Peoples Region (SNNPR) (30, 31), two from Tigri region (32, 33), and one from Afar region (34). A total of 8338 study participants (4712 males & 3626 females) were included. All studies were cross-sectional in design and the sample size of the included studies was ranged from 218 - 845 (Table 2).

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Table 2: Study characteristics included in the review and meta-analysis on knowledge about blood donation and associated factors in Ethiopia.

Author	Publication year	Region	Study participant	Method of survey	Sample size	Prevalence
Misganaw C et al.	2014	Addis Ababa	University student	Self -administered	384	83.59
Teklu S et al.	2015	Addis Ababa	Health care worker	Self –administered	295	54.24
Nigatu A & Demissie D	2014	Oromia	University student	Self -administered	399	40.35
Mulatu K et al	2017	SNNP	Community	Face to face interview	250	76.00
Gebresilase H et al.	2017	Oromia	University student	Self -administered	360	46.67
Abera B et al.	2017	Amhara	Health care worker	Self -administered	276	75.36
Mekonnen F & Melese S	2016	Amhara	Community	Face to face interview	387	56.07
Addisu AG et al.	2017	Amhara	Community	Face to face interview	376	38.30
Malako D et al.	2019	SNNP	Health care worker	Self -administered	218	82.57
Beyene GA et al.	2020	Oromia	Community	Face to face interview	410	47.07
Mijena GF et al.	2019	Oromia	Community	Self -administered	383	57.18
Jemberu YK et al.	2016	Amhara	Community	Face to face interview	772	56.48
Enawagaw B et al.	2019	Amhara	Community	Face to face interview	401	35.41
Melaku M et al.	2016	Amhara	Community	Face to face interview	768	56.77
Melaku M et al.	2018	Amhara	University student	Self -administered	255	48.24
Arage G et al.	2017	Amhara	Health care worker	Self -administered	427	51.99
Urgesa K et al.	2017	Oromia	Community	Face to face interview	845	43.55
Tadesse T et al.	2018	Tigri	Health care worker	Self -administered	556	57.19
Tadesse W et al.	2018	Afar	University student	Self -administered	339	53.98
Seid T et al.	2017	Tigri	Health care worker	Self -administered	237	70.46

SNNP, South Nation Nationalities and Peoples

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Knowledge about blood donation

The overall pooled prevalence regarding knowledge about blood donation in Ethiopia was 56.57% (95% CI: 50.30-62.84) (Figure 2).

Heterogeneity and publication bias

In this systematic review and meta-analysis, heterogeneity was detected within the studies (I2 = 97.3%, p<0.001). The funnel plot showed that there is an asymmetrical distribution of the studies included in the review and Egger's test was statistically significant (p=0.02), suggesting the presence of publication bias (Figure 3).

Subgroup analysis

In this systematic review and meta-analysis, subgroup analysis was done by region, study subjects, study setting, and sample size. The highest pooled prevalence of knowledge about blood donation was reported by health care workers, 65.28% (95% CI: 55.16, 75.40). Similarly, the level of knowledge about blood donation was higher among studies done in institutions (institution-based) 60.45% (95% CI: 51.05, 69.83) than studies done in the community. The prevalence of knowledge was reported among those studies with a sample size of <400, 60.25% (95% CI: 51.20, 6931) (Table 3).

Table 3: Subgroup analysis on the level of knowledge about blood donation and associated factors in Ethiopia (n=20)

Variables	Subgroup	Studies (n)	Population	Prevalence (95%CI)	I^2 (%)	P-value
Region	Northern Ethiopia	10	4455	54.81 (47.65, 61.57)	95.7	<0.001
	Southern Ethiopia	5	1637	58.52 (41.66,76.38)	98.2	<0.001
	Other	5	2246	58.53(42.67,74.38)	98.4	<0.001
Study participants	Health care workers	6	2009	65.28(55.16, 75.40)	95.9	<0.001
	Community	9	4592	51.84 (44.60,59.08)	96.1	<0.001
	University students	5	1737	54.62 (36.86,72.42)	95.8	<0.001
Study setting	Institution based	11	3746	60.45 (51.05, 69.83)	97.5	<0.001
	Community based	9	4592	51.84 (44.80,59.08)	96.1	<0.001
Sample size	<400	13	4159	60.25 (51.20, 6931)	97.6	<0.001
	≥ 400	7	4179	49.84 (43.90, 55.79)	93.5	<0.001

Other: Addis Ababa, Eastern Ethiopia

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Sensitivity analysis

As revealed in Table 4, none of the point's estimates outside of the overall 95% confidence interval, which confirms that the omission of any of the studies included in this systematic review and meta-analysis does not affect the overall prevalence of the level of knowledge about blood donation (Table 4).

Table 4: Sensitivity analysis of knowledge about blood donation and associated factors in Ethiopia

Study omitted	Estimate prevalence	(95%CI)
Misganaw C et al.(2014)	55.12	(49.57,60.67)
Teklu S et al.(2015)	56.69	(50.15,63.24)
Nigatu A and Demissie DB (2014)	57.42	(51.04, 63.80)
Mulatu K et al(2017)	55.55	(49.24, 6186)
Gebresilase HW et al.(2017)	57.09	(50.58, 63.59)
Abera B et al.(2017	55.58	(49.27, 61.90)
Mekonnen FH and Melese ST(2016)	56.60	(50.01,63.18)
Addisu AG et al.(2017)	57.53	(51.20, 63.86)
Malako D et al.(2019)	55.21	(49.14, 61.27)
Beyene GA et al.(2020)	57.07	(50.55, 63.59)
Mijena GF et al.(2019)	56.54	(49.95, 63.12)
Jemberu YK et al.(2016)	56.57	(49.84, 63.31)
Enawagaw B et al. (2019)	57.69	(51.48, 63.90)
Melaku M et al.(2016)	56.56	(49.83, 63.29)
Melaku M et al.(2018)	57.00	(50.50, 63.50)
Arage G et al.(2017)	56.81	(50.23, 63.39)
Urgesa K et al.(2017)	57.27	(50.82, 63.72)
Tadesse T et al.(2018)	56.54	(49.89, 63.19)
Tadesse W et al.(2018)	56.71	(50.14, 62.28)
Seid T et al.(2017)	55.85	(49.41, 62.28)
Overall	56.57	(50.30, 62.84)

CI: Confidence Interval

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Factors associated with knowledge on blood donation

This study showed that there is a significant association between sex and knowledge about blood donation. The odds of knowledge were 1.81times more likely among males than females (AOR=1.81; 95% CI; 1.44-2.28) (Figure 4). This review also showed that there is an association between the level of education and the level of knowledge about blood donation. The odds of knowledge were 3.12 times more likely among participants whose educational status was above secondary school than those who did not attend any formal education (AOR=3.12; 95% CI; 2.34-4.16) (Figure 5).

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Discussion

This systematic review and meta-analysis showed that 56.57% (95% CI: 50.30-62.84) of the population had good knowledge about blood donation. This indicates that there is a lack of awareness towards blood donation in the community. This might be due to lack of blood donation programs in formal education or curriculum, less media coverage regarding information about blood donation, limited compiegne regarding blood donation, and limited educational access in the country.

The result of this study is lower than studies conducted in Malaysia (98%) (2), Barabanki, India (90%) (35), and Benin City, Nigeria (92.65) (36). This variation might be due to the difference of the study participants in which the population of interest in the previous studies was; nursing students in Malaysia, health care workers in Benin City, Nigeria, and undergraduate medical students in Barabanki India, while this study targeted all study participants (community, health care workers, and university students). Besides, this variation might be happened due to the status and level of media coverage and educational status of the people because Ethiopia is one of the low-income countries having very limited education coverage and media accessibility. There is almost a similar finding regarding knowledge of blood donation in a study done among students of a tertiary institution in Nigeria (9). However, the prevalence of knowledge about blood donation in this study is higher than studies conducted in Kerala, India (7), and Nepal (8) in which the prevalence of knowledge was 35% and 32% respectively. It may be due to the time variation in which the previous studies were conducted before seven and five years, respectively.

In subgroup analysis, there is no significant variation regarding the prevalence of knowledge about blood donation within the region where the studies were done. However, there is a variation of knowledge prevalence within study participants, study setting, and sample size categories. Accordingly, the highest prevalence of knowledge was reported among health care workers. This is because health care workers have formal education regarding blood donation and know more about human physiology than other parts of the population. Health care workers also may have experience of blood donation. This is agreed with a study conducted in developing countries, where regular donors had good knowledge than non-donors (13). Studies conducted in institutions (institution-based) had a higher prevalence of knowledge than studies conducted in the community (community-based). This is due to the reason that populations found in the institutions had more information access (7).

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In this systematic review and meta-analysis, the odds of knowledge about blood donation were 1.81 times more likely among males than females. This might be due to the reason that females are more prone to different physiological conditions such as menstruation, bleeding during pregnancy, and delivery which deters them to donate blood. As a result, they will have poor blood donation practices when compared with males. This poor practice may indirectly affect a female's knowledge about blood donation (3). Educational status had also a significant association with the knowledge of blood donation. The odds of knowledge about blood donation were 3.12 times more likely among participants who had a level of education above secondary school than illiterates (11). This is because formal education is a cornerstone for acquiring knowledge. Therefore, participants having an educational status of above secondary level have more information accesses than participants who did not have any formal education.

Limitations of the study

All included studies in this systematic review and meta-analysis were cross-sectional studies, which may limit the generation of a causal-effect link between independent and dependent variables.

Conclusion

More than half of the study participants were knowledgeable about blood donation. Sex and educational status were significantly associated with the knowledge about blood donation in Ethiopia. Therefore, there is a need for education and dissemination of information among the general population about blood donation to build adequate knowledge and maintain a regular blood supply

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Abbreviations

AJO: African Journals Online, AOR: Adjusted Odds Ratio, CI: Confidence Interval NOS, Newcastle Ottawa Scale, PRISMA: Preferred Reporting Items for Systematic Review and Meta-analysis, and SNNP: South Nation Nationalities and Peoples.

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Declarations

Availability of data and materials

All data relevant to the study are included in the article or uploaded as supplementary information.

Author's Contributions

AG and MB designed the study, designed and run the literature search. All authors (AG, AW, MB, GG, and AD) acquired data, screened records, extracted data, and assessed the risk of bias. AG and AD did statistical analyses and wrote the report. All authors provided critical conceptual input, analyzed and interpreted the data, and critically revised the report. All authors read and approved the final manuscript.

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Competing interest

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Figure legends

1. Figure.1: Flow chart of selection for systematic review and meta-analysis on the level of knowledge about blood donation and associated factors in Ethiopia.
2. Figure 2: Forest plot of the pooled prevalence of knowledge about blood donation and associated factors in Ethiopia.
3. Figure 3: Funnel plot with 95% confidence limits of the pooled prevalence of knowledge about blood donation and associated factors in Ethiopia.
4. Figure 4: The overall pooled odds ratio of the association between sex and level of knowledge about blood donation in Ethiopia.
5. Figure 5**:** The overall pooled odds ratio of the association between educational status and level of knowledge about blood donation in Ethiopia

Additional file: Table S1: Preferred Reporting Items for Systematic Review and Meta-analysis guideline for reporting of the findings

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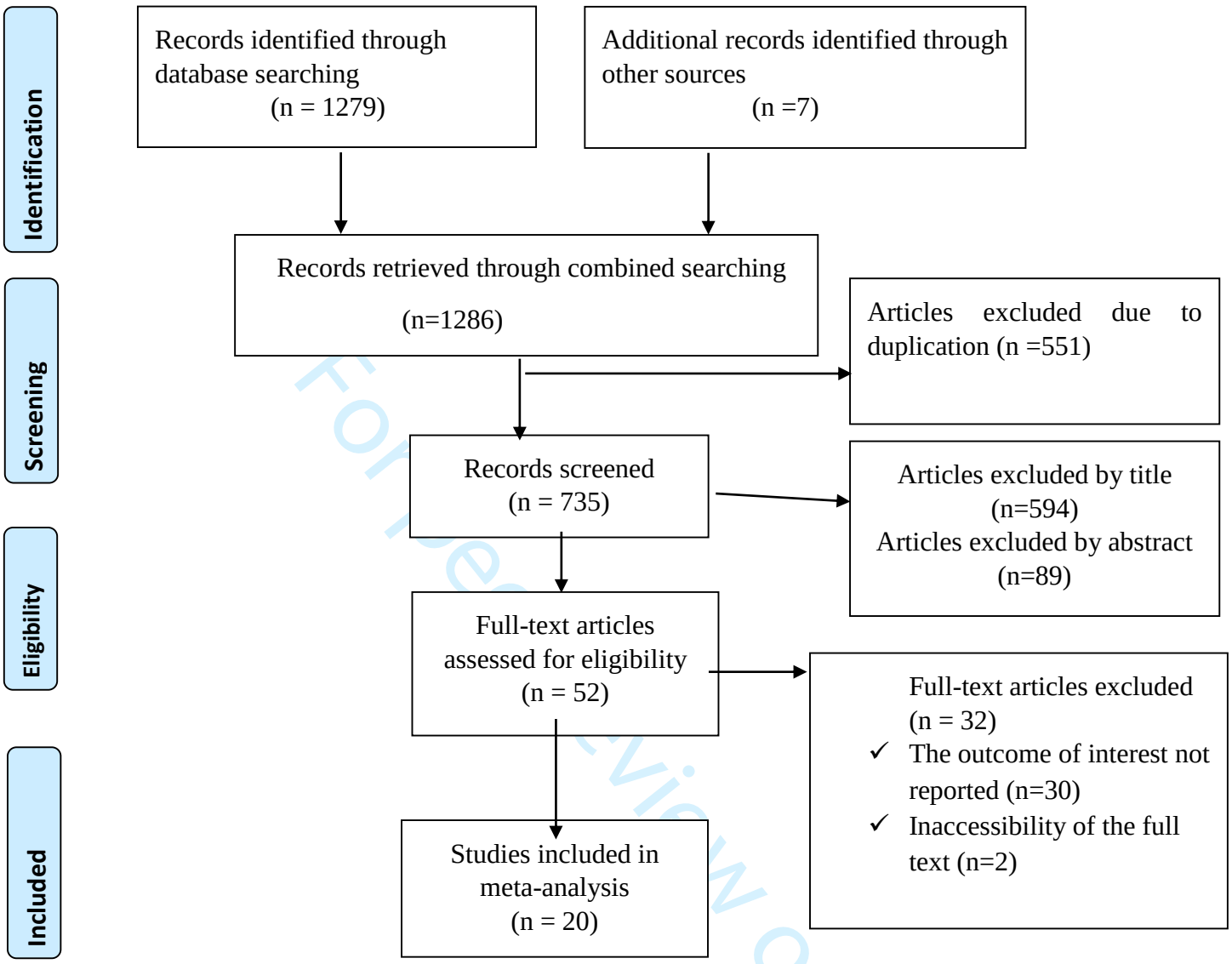


Figure.1: Flow chart of selection for systematic review and meta-analysis on the level of knowledge about blood donation and associated factors in Ethiopia.

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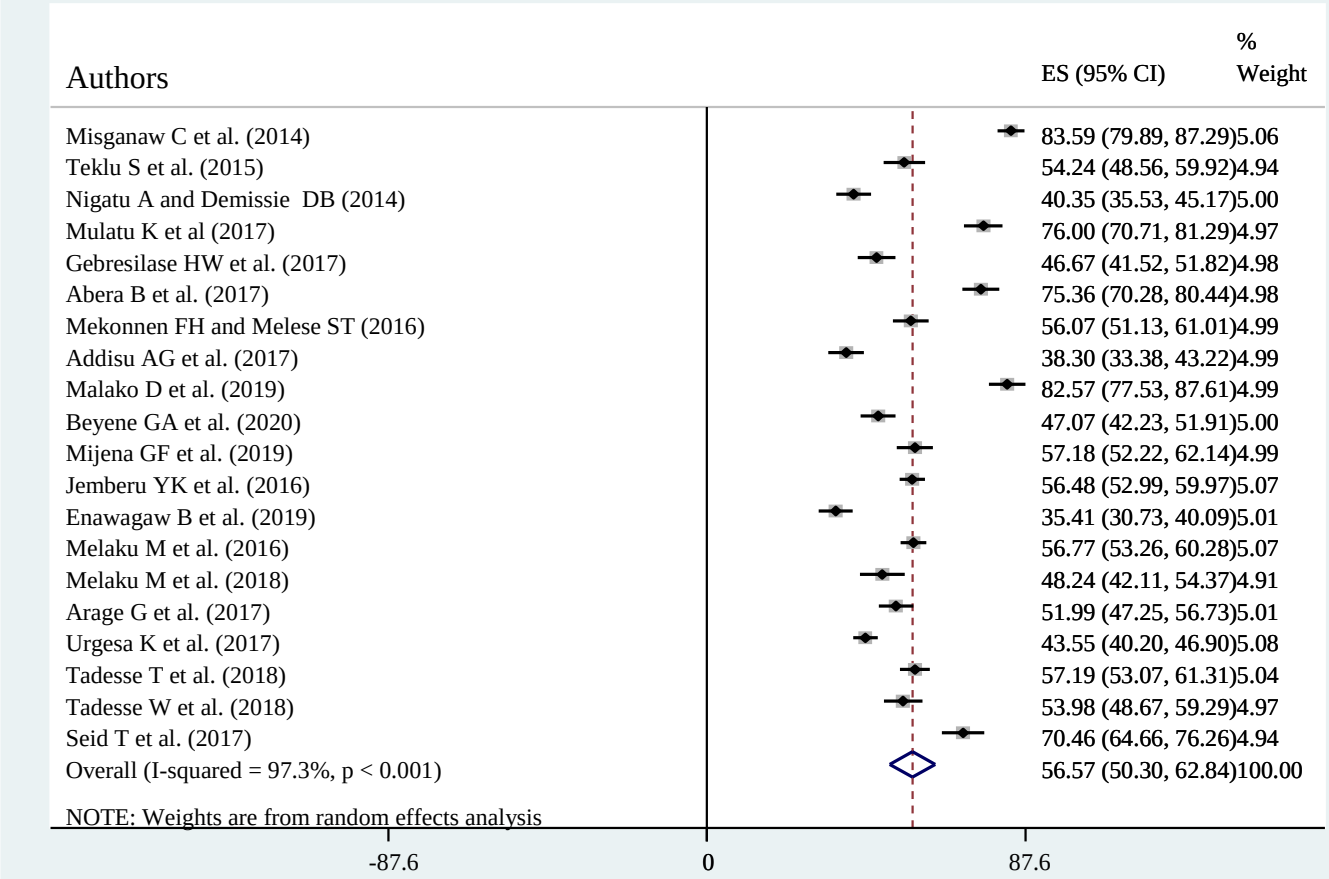


Figure 2: Forest plot of the pooled prevalence of knowledge about blood donation and associated factors in Ethiopia.

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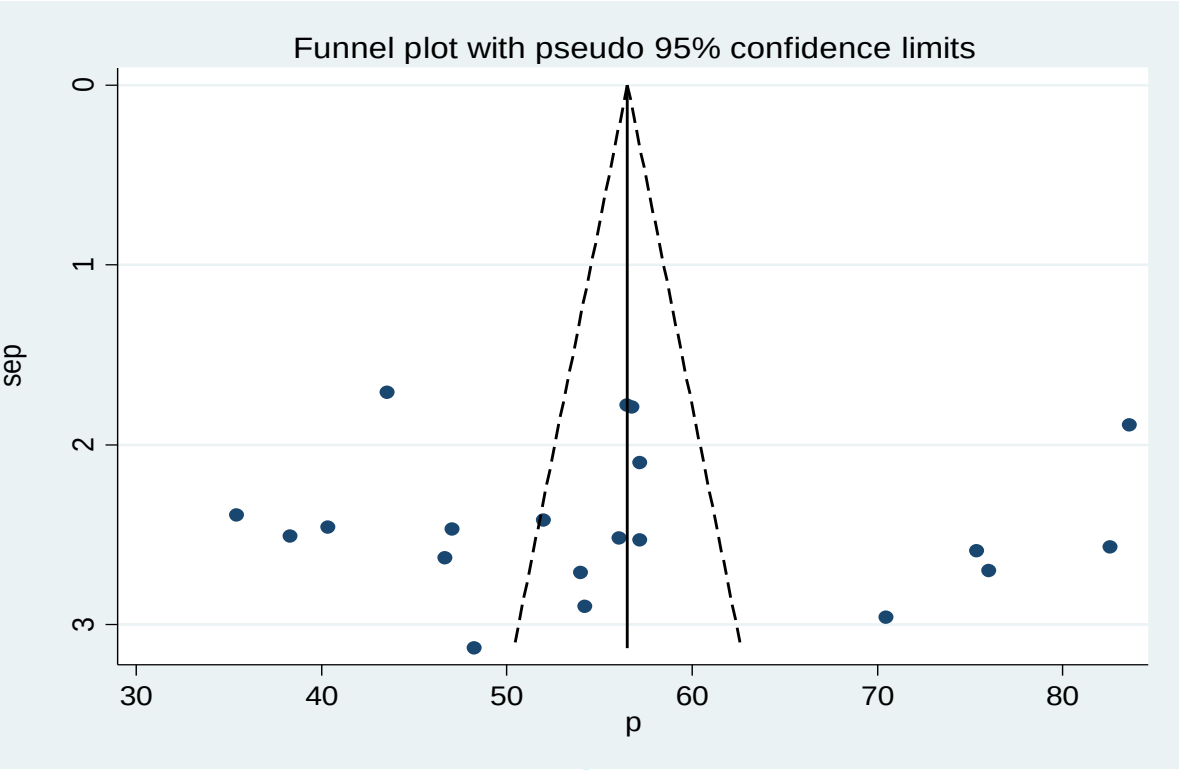


Figure 3: Funnel plot with 95% confidence limits of the pooled prevalence of knowledge about blood donation and associated factors in Ethiopia.

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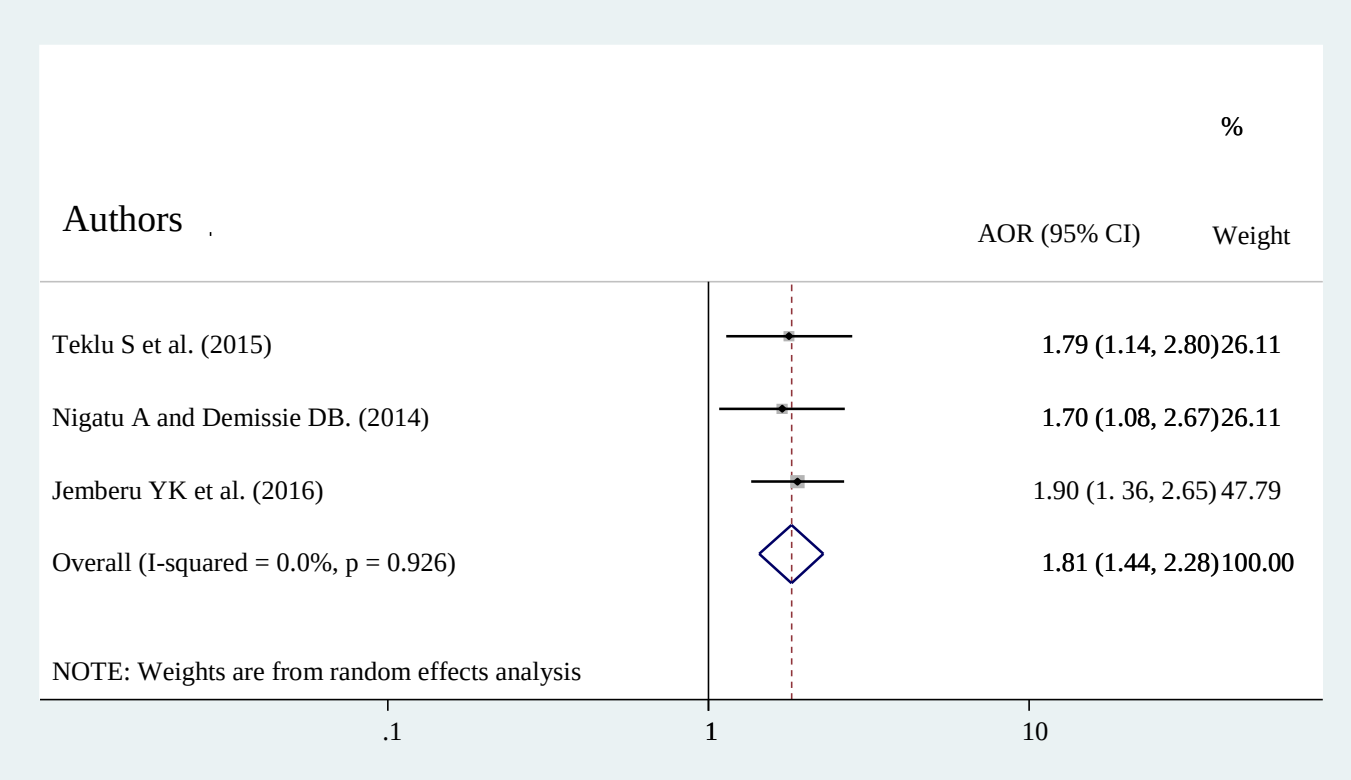


Figure 4: The overall pooled odds ratio of the association between sex and level of knowledge about blood donation in Ethiopia.

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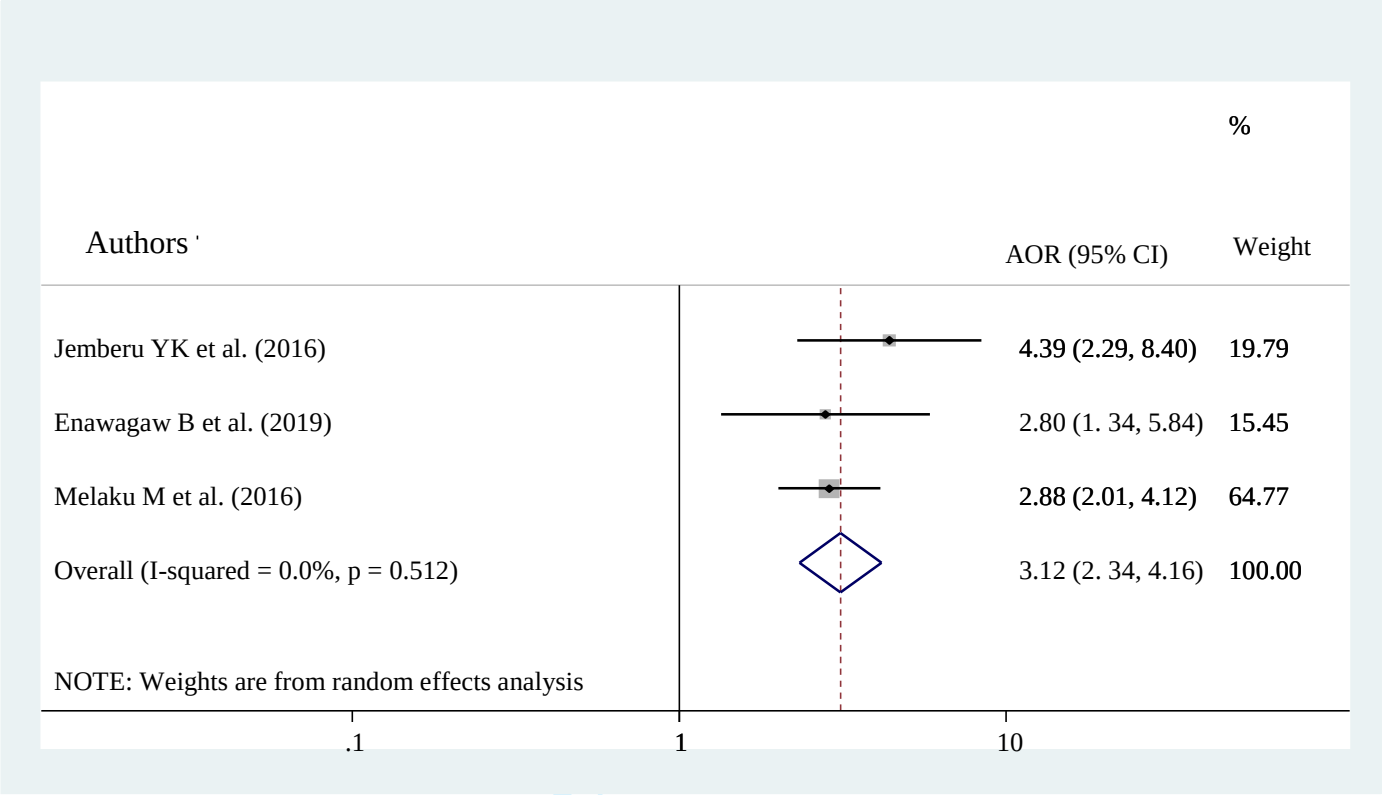


Figure 5**:** The overall pooled odds ratio of the association between educational status and level of knowledge about blood donation in Ethiopia.

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PRISMA 2009 Checklist

Table 1: Preferred Reporting Items for Systematic Review and Meta-analysis guideline for reporting of the findings

Section/topic	#	Checklist item	Reported on page #
TITLE			1
Title	1	Identify the report as a systematic review, meta-analysis, or both.	1
ABSTRACT			2
Structured summary	2	Provide a structured summary including, as applicable: background; objectives; data sources; study eligibility criteria, participants, and interventions; study appraisal and synthesis methods; results; limitations; conclusions and implications of key findings; systematic review registration number.	2
INTRODUCTION			3
Rationale	3	Describe the rationale for the review in the context of what is already known.	3
Objectives	4	Provide an explicit statement of questions being addressed with reference to participants, interventions, comparisons, outcomes, and study design (PICOS).	3
METHODS			4
Protocol and registration	5	Indicate if a review protocol exists, if and where it can be accessed (e.g., Web address), and, if available, provide registration information including registration number.	NA
Eligibility criteria	6	Specify study characteristics (e.g., PICOS, length of follow-up) and report characteristics (e.g., years considered, language, publication status) used as criteria for eligibility, giving rationale.	5
Information sources	7	Describe all information sources (e.g., databases with dates of coverage, contact with study authors to identify additional studies) in the search and date last searched.	4
Search	8	Present full electronic search strategy for at least one database, including any limits used, such that it could be repeated.	4
Study selection	9	State the process for selecting studies (i.e., screening, eligibility, included in systematic review, and, if applicable, included in the meta-analysis).	5
Data collection process	10	Describe method of data extraction from reports (e.g., piloted forms, independently, in duplicate) and any processes for obtaining and confirming data from investigators.	5
Data items	11	List and define all variables for which data were sought (e.g., PICOS, funding sources) and any assumptions and simplifications made.	5
Risk of bias in individual studies	12	Describe methods used for assessing risk of bias of individual studies (including specification of whether this was done at the study or outcome level), and how this information is to be used in any data synthesis.	6
Summary measures	13	State the principal summary measures (e.g., risk ratio, difference in means).	6
Synthesis of results	14	Describe the methods of handling data and combining results of studies, if done, including measures of consistency (e.g., I^2) for each meta-analysis.	6

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PRISMA 2009 Checklist

Section/topic	#	Checklist item	Reported on page #
Risk of bias across studies	15	Specify any assessment of risk of bias that may affect the cumulative evidence (e.g., publication bias, selective reporting within studies).	6
Additional analyses	16	Describe methods of additional analyses (e.g., sensitivity or subgroup analyses, meta-regression), if done, indicating which were pre-specified.	6
RESULTS			7
Study selection	17	Give numbers of studies screened, assessed for eligibility, and included in the review, with reasons for exclusions at each stage, ideally with a flow diagram.	7
Study characteristics	18	For each study, present characteristics for which data were extracted (e.g., study size, PICOS, follow-up period) and provide the citations.	8
Risk of bias within studies	19	Present data on risk of bias of each study and, if available, any outcome level assessment (see item 12).	10
Results of individual studies	20	For all outcomes considered (benefits or harms), present, for each study: (a) simple summary data for each intervention group (b) effect estimates and confidence intervals, ideally with a forest plot.	9
Synthesis of results	21	Present results of each meta-analysis done, including confidence intervals and measures of consistency.	10
Risk of bias across studies	22	Present results of any assessment of risk of bias across studies (see Item 15).	11
Additional analysis	23	Give results of additional analyses, if done (e.g., sensitivity or subgroup analyses, meta-regression [see Item 16]).	12
DISCUSSION			14
Summary of evidence	24	Summarize the main findings including the strength of evidence for each main outcome; consider their relevance to key groups (e.g., healthcare providers, users, and policy makers).	14
Limitations	25	Discuss limitations at study and outcome level (e.g., risk of bias), and at review-level (e.g., incomplete retrieval of identified research, reporting bias).	16
Conclusions	26	Provide a general interpretation of the results in the context of other evidence, and implications for future research.	16
FUNDING			NA
Funding	27	Describe sources of funding for the systematic review and other support (e.g., supply of data); role of funders for the systematic review.	NA

From: Moher D, Liberati A, Tetzlaff J, Altman DG, The PRISMA Group (2009). Preferred Reporting Items for Systematic Reviews and Meta-Analyses: The PRISMA Statement. PLoS Med 6(7): e1000097. doi:10.1371/journal.pmed1000097

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Knowledge on blood donation and associated factors in Ethiopia: A systematic review and Meta-analysis

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Abstract

Objective: To assess the level of knowledge about blood donation and associated factors in Ethiopia.

Design: Systematic review and meta-analysis

Methods: Both published and unpublished cross-sectional studies conducted on the level of knowledge about blood donation in Ethiopia were included. Articles from different databases such as PubMed/MEDLINE, HINARI, EMBASE, Scopus, Google Scholar, and African Journals Online (AJOL) were searched. Cochrane I^2 statistics were used to check the existence of heterogeneity. Subgroup analysis and sensitivity analysis were carried out on evidence of heterogeneity. Egger test with funnel plot was conducted to investigate publication bias.

Result: Twenty cross-sectional studies with a total of 8,338 study participants (4,712 males & 3,626 females) were involved. The overall nationwide level of knowledge about blood donation was 56.57% (95% CI: 50.30-62.84). Being a secondary school and above (AOR=3.12; 95% CI; 2.34-4.16) and being male (AOR=1.81; 95% CI; 1.44-2.28) were factors associated with a level of knowledge about blood donation.

Conclusion: More than half of the study participants were knowledgeable about blood donation. Sex and educational status were significantly associated factors with a level of knowledge about blood donation in Ethiopia. Therefore, there is a need for education and dissemination of information among the general population about blood donation to build adequate knowledge and maintain a regular blood supply.

Keywords: *Knowledge, blood donation practice, and associated factors*

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Strength and weakness of the study

- This study highlights the nationwide level of knowledge towards blood donation in Ethiopia
- This study covers a wide area and accuses different articles, making the review more accurate
- Subgroup analysis and sensitivity analysis were carried out to investigate the heterogeneity of the included studies.
- All included studies were cross-sectional in design, which may limit to show the cause-effect relationship.

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Introduction

Blood donation is a lifesaving practice for people who face different emergency conditions like road traffic accidents, surgical procedures, pregnancy and delivery complications, chemotherapy, and diseases such as malaria, anemia, and intestinal parasites(1). Saving the lives of peoples through blood donation is a noble practice to save the lives of millions. People can save up to three lives of patients by donating safe and adequate blood, which is fundamental for the survival of patients who need blood (2).

Although the demand for adequate and safe blood is significantly increasing, there is a serious and critical shortage of blood in stocks from low and middle-income countries. In low and middle-income countries, family replacement and paid blood donors are the commonest sources of blood transfusion (3). The availability of blood for transfusion in low-income countries is too limited due to the low level of knowledge, unwillingness to donate blood, and poor blood donation practice. In the meanwhile, patients are suffering from a lack of blood transfusion due to the increment of different medical and surgical conditions (4).

Even if every blood donor is considered a hero, the amount of blood collected from blood donors and the average blood collection rate in low-income countries including Ethiopia is low compared to the demand for blood (5). Ignorance of the population having basic knowledge and acts of replacement therapy other than collecting from volunteer blood donors were predisposing factors to have a misconception towards blood donation practice in the world (6).

As different studies showed, the prevalence of level of knowledge regarding blood donation was ranged from 32.4%-40.45% (7), (4), (8). Education, sensitization regarding blood donation, availing of public awareness, Compiegne through the internet and media are the suggested strategies to increase the awareness, attitude, and motivational practice of blood donation (9-11). Fear of different health risks after donating blood and lack of information on where, when, and how to donate blood are the commonest hindering factors towards blood donation practice (12). Even though beliefs, attitudes, and behaviors regarding blood donation differ, adequate knowledge is needed among non-donors. There is a gap between willingness to blood donation and the number of donors in most populations around the world, which affects blood donation practice (13). There is a need to spread awareness among the general population about blood donation to maintain an adequate and safe blood supply through well-designed communication strategies to overcome the problem (7).

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Although the need for blood supply is too high in Ethiopia, there is a serious shortage of blood. Thus, identifying different factors that hinder the knowledge of blood donation practice is essential. Therefore, this systematic review and meta-analysis aim to review different studies conducted so far on the level of knowledge about blood donation and associated factors in Ethiopia.

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Methods and materials

Study protocol

In this meta-analysis, the Preferred Reporting Items for Systematic Review and Meta-analysis (PRISMA) guideline for reporting the findings was used (14) (Table S1).

Databases and search strategy

Different databases such as PubMed/MEDLINE, HINARI, EMBASE, Google Scholar, and African Journal Online (AJOL) were used to search available articles. Both Published and unpublished articles from the repository of Ethiopian universities were also searched. The searching date was from January 1st, 2000 up to May 01, 2021. Articles reporting on knowledge of blood donation and associated factors in Ethiopia were included in the final analysis. The search items were; "knowledge" OR "awareness" AND "blood donation" OR "volunteer blood donation" AND "associated factors" OR "determinant factors" AND "Ethiopia". These search strings were developed using "AND" and "OR" Boolean operators **(**Table 1**).**

Table 1: Searches from different databases about the level of knowledge and associated factors regarding blood donation in Ethiopia.

Databasis	Searching terms	Number of studies
MEDLINE/ PubMed	"knowledge" OR "awareness "AND "blood donation" OR "volunteer blood donation "AND "associated factors" OR "determinant factors" AND "Ethiopia".	235
Google Scholar	"knowledge" OR "awareness "AND "blood donation" OR "volunteer blood donation "AND "associated factors" OR "determinant factors" AND "Ethiopia".	352
Other databasis		7
Total retrieved articles		694
Included studies		20

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Searching and eligibility of studies

All retrieved articles were exported into "EndNote reference software version 8 (Thomson, Stamford, CT, USA) citation manager" to sort and avoid duplication. Two investigators (AG and AD) independently evaluated each article by title and abstract, assessed the eligibility of the articles, and reviewed the selected articles critically. Extraction was done by the authors' name, publication year, a region where the study was conducted, study subjects, sampling method, method of survey, study period, sample size, study design, level of education, level of knowledge regarding blood donation, and factors associated with the level of knowledge about blood donation.

Eligibility criteria

Both published and unpublished cross-sectional studies conducted on the level of knowledge about blood donation among populations in Ethiopia were included, whereas qualitative studies, different trials, case reports, review articles, updates, and news were excluded from the analysis. Furthermore, articles not reporting the outcome of the study and those without full text were excluded.

Outcome measurement of the study

There are two main outcomes such as the level of knowledge about blood donation and factors associated with it. The level of knowledge was measured by using the mean score of knowledge assessment questions. Thus, those study participants who scored mean and above on the knowledge assessment items were considered as knowledgeable, whereas those who scored below the mean score were considered as not knowledgeable about blood donation.

Quality Assessment

Two authors (AG and AD) independently assessed the quality of the studies using the Newcastle Ottawa Scale (NOS) for cross-sectional studies (15). Methodological quality, comparability, and outcome and statistical analysis of the studies were the main assessment tools used to declare the quality of the studies. Studies that scored a scale of ≥ 7 out of 10 were considered as achieving high quality. During quality appraisal of the articles, any discrepancies between the two authors were resolved. All authors independently assessed for the eligibility of articles to be considered in the final analysis.

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Data processing and analysis

In this systematic review and meta-analysis, a weighted inverse variance random-effects model at 95% CI was used to calculate the pooled prevalence of level of knowledge and associated factors about blood donation (16). After extraction and cleaning using Microsoft Excel spreadsheets, the data were exported to STATA version 11 statistical software for analysis. The heterogeneity of the studies was assessed using the Cochrane Q-test and I^2 with its corresponding p-value. The values of I^2: 25%, 50%, and 75% represent low, moderate, and high heterogeneity, respectively (17). The source of heterogeneity was examined through subgroup analysis based on (region, study subjects, study setting, and sample size). Sensitivity analysis was also carried out to confirm the presence or absence of influential studies. The presence of publication bias was evaluated by using Egger's test and presented with funnel plots (18). For associated factors, a log odds ratio was used to decide the association between associated factors and the level of knowledge about blood donation. A statistical test with a P value of less than 0.05 was considered statistically significant (19).

Patient and public involvement

It was not appropriate or possible to involve patients or the public in the design, conduct, reporting, or dissemination plans of our research

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Result

Six hundred ninety-four articles were retrieved. Of these retrieved articles, 284 were excluded due to duplication. Three hundred sixty-eight articles were further excluded after reviewing titles and abstracts. Furthermore, 22 articles were excluded which did not fulfill the inclusion criteria. Then, 20 articles were used for the final analysis (Figure 1).

Characteristics of the studies and study participants

Twenty cross-sectional studies with a total of 8,338 study participants (4,712 males & 3,626 females) were involved. Of the included articles, eight were from the Amhara region (1, 12, 20-25), five were from Oromia region (4, 26-29), two were from Addis Ababa city administration (5, 30), two were from Southern Nation Nationalities and Peoples Region (SNNPR) (31, 32), two were from Tigri region (33, 34), and the remaining one was from Afar region (35). The sample size of the included studies was ranged from 218 - 845 (Table 2).

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Table 2: Study characteristics included in the review and meta-analysis on knowledge about blood donation and associated factors in Ethiopia.

Author	Publication year	Region	Study participant	Method of survey	Sample size	Prevalence
Misganaw C et al.	2014	Addis Ababa	University student	Self -administered	384	83.59
Teklu S et al.	2015	Addis Ababa	Health care worker	Self –administered	295	54.24
Nigatu A & Demissie D	2014	Oromia	University student	Self -administered	399	40.35
Mulatu K et al	2017	SNNP	Community	Face to face interview	250	76.00
Gebresilase H et al.	2017	Oromia	University student	Self -administered	360	46.67
Abera B et al.	2017	Amhara	Health care worker	Self -administered	276	75.36
Mekonnen F & Melese S	2016	Amhara	Community	Face to face interview	387	56.07
Addisu AG et al.	2017	Amhara	Community	Face to face interview	376	38.30
Malako D et al.	2019	SNNP	Health care worker	Self -administered	218	82.57
Beyene GA et al.	2020	Oromia	Community	Face to face interview	410	47.07
Mijena GF et al.	2019	Oromia	Community	Self -administered	383	57.18
Jemberu YK et al.	2016	Amhara	Community	Face to face interview	772	56.48
Enawgaw B et al.	2019	Amhara	Community	Face to face interview	401	35.41
Melaku M et al.	2016	Amhara	Community	Face to face interview	768	56.77
Melaku M et al.	2018	Amhara	University student	Self -administered	255	48.24
Arage G et al.	2017	Amhara	Health care worker	Self -administered	427	51.99
Urgesa K et al.	2017	Oromia	Community	Face to face interview	845	43.55
Tadesse T et al.	2018	Tigri	Health care worker	Self -administered	556	57.19
Tadesse W et al.	2018	Afar	University student	Self -administered	339	53.98
Seid T et al.	2017	Tigri	Health care worker	Self -administered	237	70.46

SNNP, South Nation Nationalities and Peoples

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Knowledge about blood donation

The overall pooled prevalence of level of knowledge about blood donation in Ethiopia was 56.57% (95% CI: 50.30-62.84) (Figure 2).

Heterogeneity and publication bias

In this systematic review and meta-analysis, heterogeneity was detected within the studies (I^2 = 97.3%, p<0.001). The funnel plot showed that there is an asymmetrical distribution of the studies included in the review and Egger's test was statistically significant (p=0.02), suggesting the presence of publication bias (Figure 3).

Subgroup analysis

Subgroup analysis was done by region, study subjects, study setting, and sample size. The highest pooled prevalence of level of knowledge about blood donation was reported by health care workers, 65.28% (95% CI: 55.16, 75.40). Similarly, the level of knowledge about blood donation was higher among studies done in institutions (institution-based) 60.45% (95% CI: 51.05, 69.83) than studies conducted in the community (Table 3).

Table 3: Subgroup analysis on the level of knowledge about blood donation and associated factors in Ethiopia (n=20)

Variables	Subgroup	Studies (n)	Population	Prevalence (95%CI)	I^2 (%)	P-value
Region	Northern Ethiopia	10	4455	54.81 (47.65, 61.57)	95.7	<0.001
	Southern Ethiopia	5	1637	58.52 (41.66,76.38)	98.2	<0.001
	Other	5	2246	58.53(42.67,74.38)	98.4	<0.001
Study participants	Health care workers	6	2009	65.28(55.16, 75.40)	95.9	<0.001
	Community	9	4592	51.84 (44.60,59.08)	96.1	<0.001
	University students	5	1737	54.62 (36.86,72.42)	95.8	<0.001
Study setting	Institution based	11	3746	60.45 (51.05, 69.83)	97.5	<0.001
	Community based	9	4592	51.84 (44.80,59.08)	96.1	<0.001
Sample size	<400	13	4159	60.25 (51.20, 6931)	97.6	<0.001
	≥ 400	7	4179	49.84 (43.90, 55.79)	93.5	<0.001

Other: Addis Ababa, Eastern Ethiopia

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Sensitivity analysis

As revealed in Table 4, none of the point's estimates outside of the overall 95% confidence interval, which confirms that the omission of any of the studies included in this systematic review and meta-analysis does not affect the overall prevalence of the level of knowledge about blood donation (Table 4).

Table 4: Sensitivity analysis of knowledge about blood donation and associated factors in Ethiopia

Study omitted	Estimate prevalence	(95%CI)
Misganaw C et al.(2014)	55.12	(49.57,60.67)
Teklu S et al.(2015)	56.69	(50.15,63.24)
Nigatu A and Demissie DB (2014)	57.42	(51.04, 63.80)
Mulatu K et al(2017)	55.55	(49.24, 6186)
Gebresilase HW et al.(2017)	57.09	(50.58, 63.59)
Abera B et al.(2017	55.58	(49.27, 61.90)
Mekonnen FH and Melese ST(2016)	56.60	(50.01,63.18)
Addisu AG et al.(2017)	57.53	(51.20, 63.86)
Malako D et al.(2019)	55.21	(49.14, 61.27)
Beyene GA et al.(2020)	57.07	(50.55, 63.59)
Mijena GF et al.(2019)	56.54	(49.95, 63.12)
Jemberu YK et al.(2016)	56.57	(49.84, 63.31)
Enawgaw B et al. (2019)	57.69	(51.48, 63.90)
Melaku M et al.(2016)	56.56	(49.83, 63.29)
Melaku M et al.(2018)	57.00	(50.50, 63.50)
Arage G et al.(2017)	56.81	(50.23, 63.39)
Urgesa K et al.(2017)	57.27	(50.82, 63.72)
Tadesse T et al.(2018)	56.54	(49.89, 63.19)
Tadesse W et al.(2018)	56.71	(50.14, 62.28)
Seid T et al.(2017)	55.85	(49.41, 62.28)
Overall	56.57	(50.30, 62.84)

CI: Confidence Interval

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Factors associated with knowledge on blood donation

In this study, the sex of participants and level of education were significant factors associated with the level of knowledge about blood donation. Thus, the odds of the level of knowledge were 1.81times more likely among males than females (AOR=1.81; 95% CI; 1.44-2.28) (Figure 4). Similarly, the odds of the level of knowledge were 3.12 times more likely among participants whose educational status was above secondary school than those who did not attend any formal education (AOR=3.12; 95% CI; 2.34-4.16) (Figure 5).

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Discussion

In this systematic review and meta-analysis, the pooled prevalence of good knowledge about blood donation is 56.57% (95% CI: 50.30-62.84), which indicates that there is a lack of adequate knowledge about blood donation in the country. This might be due to the absence of regular blood donation programs, less media coverage, limited Compiegne, and limited educational access regarding blood donation in the country.

The result of this study is lower than studies conducted in Malaysia (98%) (2), Barabanki, India (90%) (36), and Benin City, Nigeria (92.65 %) (37). This variation might be due to the difference in the study participants. The study participants were nursing students in Malaysia, health care workers in Benin City, Nigeria, and undergraduate medical students in Barabanki, India. However, this study targeted all populations in the country. Besides, this variation might be happened due to the status and level of media coverage and educational status of the people because Ethiopia is one of the low-income countries having very limited education coverage and media accessibility. There is almost a similar finding regarding knowledge of blood donation in a study done among students of a tertiary institution in Nigeria (9). However, the finding of this study is higher than studies conducted in Kerala, India (35%) (7), and Nepal (32%) (8). It may be due to the time variation in which the previous studies were conducted before seven and five years, respectively.

In the subgroup analysis, there is no significant variation regarding the prevalence of knowledge about blood donation within the region where the studies were done. However, there were variations regarding knowledge prevalence within study participants, study setting, and sample size categories. Accordingly, the highest prevalence of knowledge was reported among health care workers. This is because health care workers have formal education regarding blood donation and know more about human physiology than others. Health care workers also may have experience with blood donation. This is agreed with a study conducted in developing countries, where regular donors had good knowledge than non-donors (13). Studies conducted in institutions (institution-based) had a higher prevalence of knowledge than studies conducted in the community (community-based). This is due to the reason that populations found in the institutions had more information access (7).

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In this systematic review and meta-analysis, the odds of knowledge about blood donation were 1.81 times more likely among males than females. This might be due to the reason that females are more prone to different physiological conditions such as menstruation, bleeding during pregnancy, and delivery which deters them to donate blood. As a result, they will have poor blood donation practices when compared with males. This poor practice may indirectly affect a female's knowledge about blood donation (3). Educational status had also a significant association with the knowledge of blood donation. The odds of knowledge about blood donation were 3.12 times more likely among participants who had a level of education above secondary school than illiterates (11). This is because formal education is a cornerstone for acquiring knowledge. Therefore, participants having an educational status of being a secondary school and above have more information access than participants who did not have any formal education.

Limitations of the study

All included studies in this systematic review and meta-analysis were cross-sectional studies, which may limit the generation of a causal-effect link between independent and dependent variables.

Conclusion

More than half of the study participants were knowledgeable about blood donation. Sex and educational status were significantly associated with the level of knowledge about blood donation in Ethiopia. Therefore, there is a need for education and dissemination of information among the general population about blood donation to build adequate knowledge and maintain a regular blood supply

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Abbreviations

AJO: African Journals Online, AOR: Adjusted Odds Ratio, CI: Confidence Interval NOS, Newcastle Ottawa Scale, PRISMA: Preferred Reporting Items for Systematic Review and Meta-analysis, and SNNP: South Nation Nationalities and Peoples.

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Declarations

Availability of data and materials

All data relevant to the study are included in the article or uploaded as supplementary information.

Author's Contributions

AG and MB designed the study, designed and run the literature search. All authors (AG, AW, MB, GG, and AD) acquired data, screened records, extracted data, and assessed the risk of bias. AG and AD did statistical analyses and wrote the report. All authors provided critical conceptual input, analyzed and interpreted the data, and critically revised the report. All authors read and approved the final manuscript.

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Figure legends

1. Figure.1: Flow chart of selection for systematic review and meta-analysis on the level of knowledge about blood donation and associated factors in Ethiopia.
2. Figure 2: Forest plot of the pooled prevalence of knowledge about blood donation and associated factors in Ethiopia.
3. Figure 3: Funnel plot with 95% confidence limits of the pooled prevalence of knowledge about blood donation and associated factors in Ethiopia.
4. Figure 4: The overall pooled odds ratio of the association between sex and level of knowledge about blood donation in Ethiopia.
5. Figure 5**:** The overall pooled odds ratio of the association between educational status and level of knowledge about blood donation in Ethiopia

Additional file: Table S1: Preferred Reporting Items for Systematic Review and Meta-analysis guideline for reporting of the findings

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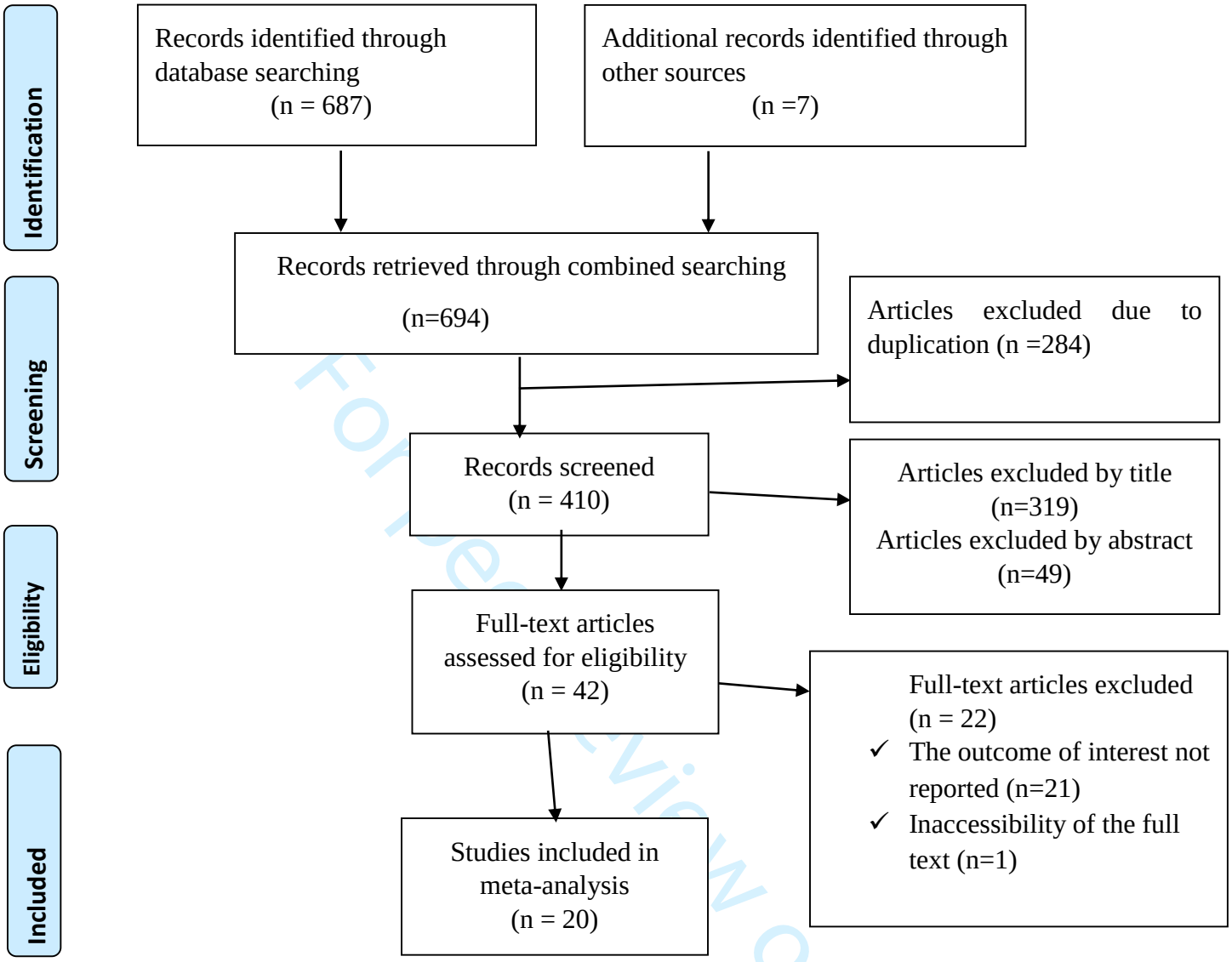


Figure.1: Flow chart of selection for systematic review and meta-analysis on the level of knowledge about blood donation and associated factors in Ethiopia.

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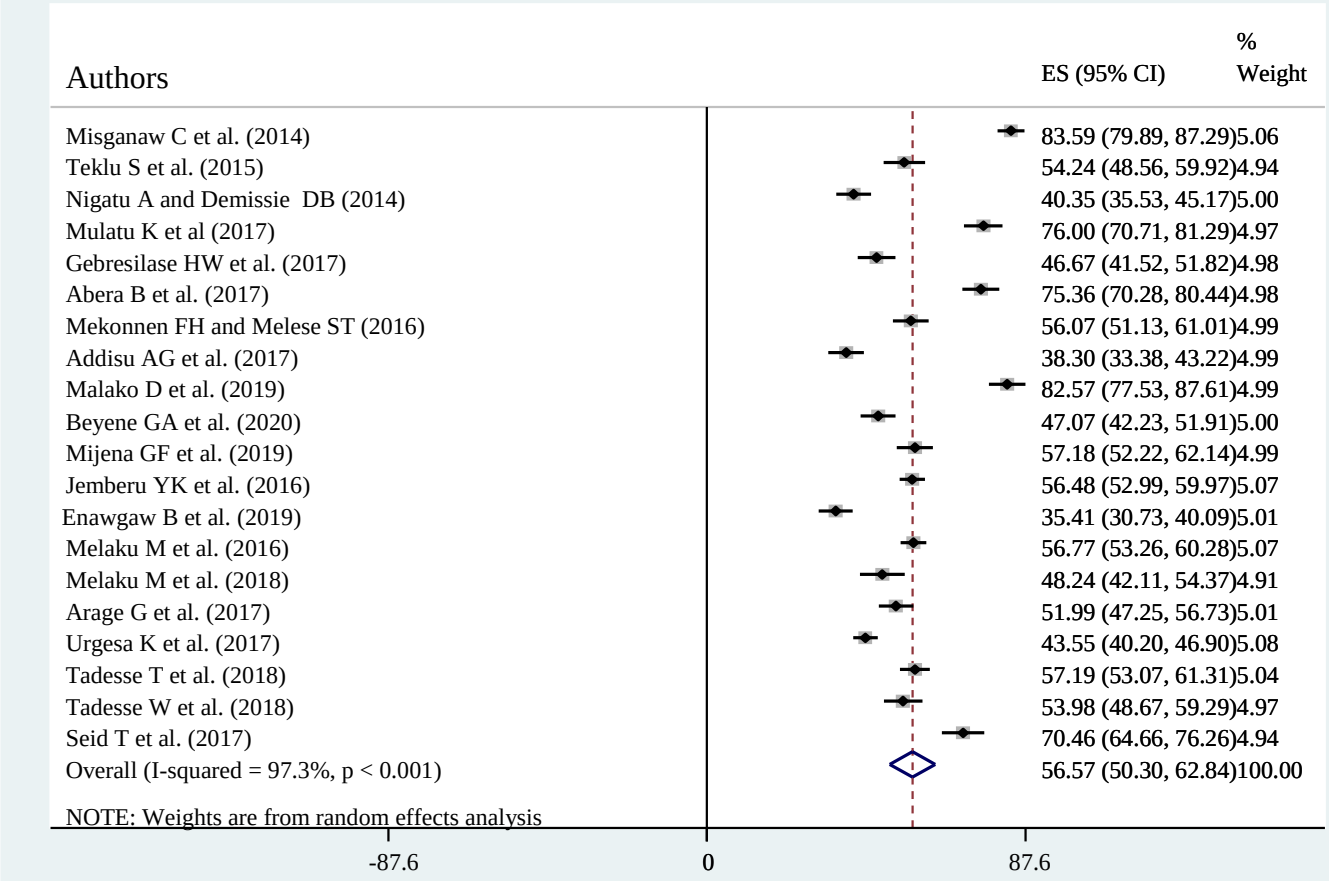


Figure 2: Forest plot of the pooled prevalence of knowledge about blood donation and associated factors in Ethiopia.

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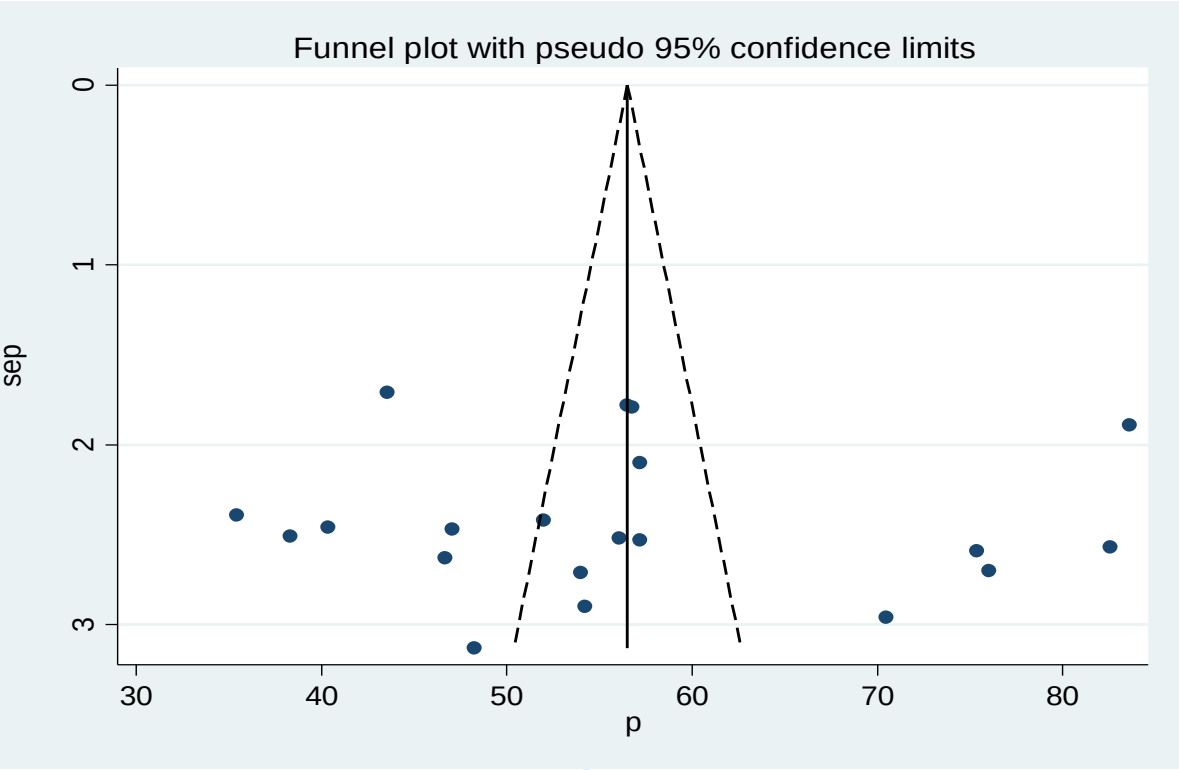


Figure 3: Funnel plot with 95% confidence limits of the pooled prevalence of knowledge about blood donation and associated factors in Ethiopia.

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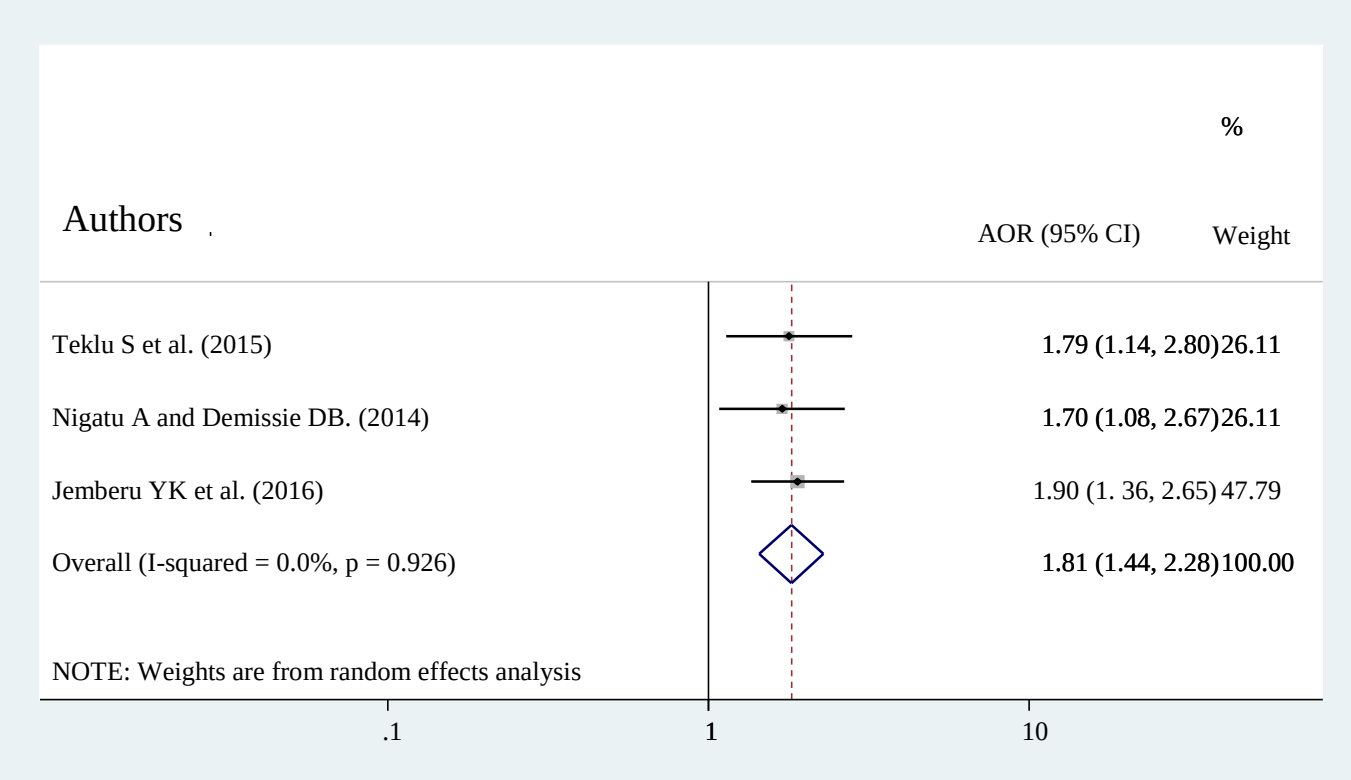


Figure 4: The overall pooled odds ratio of the association between sex and level of knowledge about blood donation in Ethiopia.

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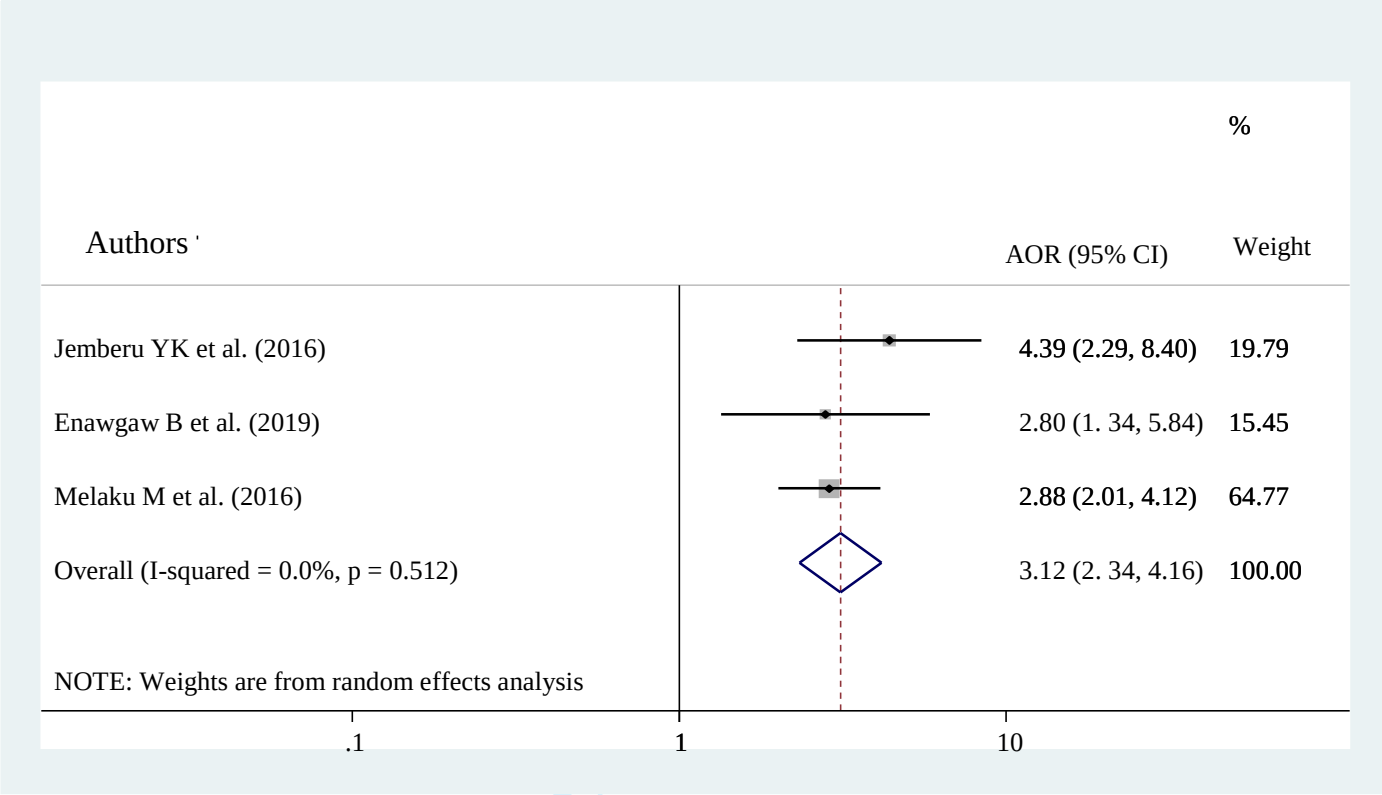


Figure 5: The overall pooled odds ratio of the association between educational status and level of knowledge about blood donation in Ethiopia.

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PRISMA 2009 Checklist

Table 1: Preferred Reporting Items for Systematic Review and Meta-analysis guideline for reporting of the findings

Section/topic	#	Checklist item	Reported on page #
TITLE			1
Title	1	Identify the report as a systematic review, meta-analysis, or both.	1
ABSTRACT			2
Structured summary	2	Provide a structured summary including, as applicable: background; objectives; data sources; study eligibility criteria, participants, and interventions; study appraisal and synthesis methods; results; limitations; conclusions and implications of key findings; systematic review registration number.	2
INTRODUCTION			3
Rationale	3	Describe the rationale for the review in the context of what is already known.	3
Objectives	4	Provide an explicit statement of questions being addressed with reference to participants, interventions, comparisons, outcomes, and study design (PICOS).	3
METHODS			4
Protocol and registration	5	Indicate if a review protocol exists, if and where it can be accessed (e.g., Web address), and, if available, provide registration information including registration number.	NA
Eligibility criteria	6	Specify study characteristics (e.g., PICOS, length of follow-up) and report characteristics (e.g., years considered, language, publication status) used as criteria for eligibility, giving rationale.	5
Information sources	7	Describe all information sources (e.g., databases with dates of coverage, contact with study authors to identify additional studies) in the search and date last searched.	4
Search	8	Present full electronic search strategy for at least one database, including any limits used, such that it could be repeated.	4
Study selection	9	State the process for selecting studies (i.e., screening, eligibility, included in systematic review, and, if applicable, included in the meta-analysis).	5
Data collection process	10	Describe method of data extraction from reports (e.g., piloted forms, independently, in duplicate) and any processes for obtaining and confirming data from investigators.	5
Data items	11	List and define all variables for which data were sought (e.g., PICOS, funding sources) and any assumptions and simplifications made.	5
Risk of bias in individual studies	12	Describe methods used for assessing risk of bias of individual studies (including specification of whether this was done at the study or outcome level), and how this information is to be used in any data synthesis.	6
Summary measures	13	State the principal summary measures (e.g., risk ratio, difference in means).	6
Synthesis of results	14	Describe the methods of handling data and combining results of studies, if done, including measures of consistency (e.g., I^2) for each meta-analysis.	6

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PRISMA 2009 Checklist

Section/topic	#	Checklist item	Reported on page #
Risk of bias across studies	15	Specify any assessment of risk of bias that may affect the cumulative evidence (e.g., publication bias, selective reporting within studies).	6
Additional analyses	16	Describe methods of additional analyses (e.g., sensitivity or subgroup analyses, meta-regression), if done, indicating which were pre-specified.	6
RESULTS			7
Study selection	17	Give numbers of studies screened, assessed for eligibility, and included in the review, with reasons for exclusions at each stage, ideally with a flow diagram.	7
Study characteristics	18	For each study, present characteristics for which data were extracted (e.g., study size, PICOS, follow-up period) and provide the citations.	8
Risk of bias within studies	19	Present data on risk of bias of each study and, if available, any outcome level assessment (see item 12).	10
Results of individual studies	20	For all outcomes considered (benefits or harms), present, for each study: (a) simple summary data for each intervention group (b) effect estimates and confidence intervals, ideally with a forest plot.	9
Synthesis of results	21	Present results of each meta-analysis done, including confidence intervals and measures of consistency.	10
Risk of bias across studies	22	Present results of any assessment of risk of bias across studies (see Item 15).	11
Additional analysis	23	Give results of additional analyses, if done (e.g., sensitivity or subgroup analyses, meta-regression [see Item 16]).	12
DISCUSSION			14
Summary of evidence	24	Summarize the main findings including the strength of evidence for each main outcome; consider their relevance to key groups (e.g., healthcare providers, users, and policy makers).	14
Limitations	25	Discuss limitations at study and outcome level (e.g., risk of bias), and at review-level (e.g., incomplete retrieval of identified research, reporting bias).	16
Conclusions	26	Provide a general interpretation of the results in the context of other evidence, and implications for future research.	16
FUNDING			NA
Funding	27	Describe sources of funding for the systematic review and other support (e.g., supply of data); role of funders for the systematic review.	NA

From: Moher D, Liberati A, Tetzlaff J, Altman DG, The PRISMA Group (2009). Preferred Reporting Items for Systematic Reviews and Meta-Analyses: The PRISMA Statement. PLoS Med 6(7): e1000097. doi:10.1371/journal.pmed1000097

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